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Inventor(s)

SEO; Satoshi et al.

LIGHT-EMITTING DEVICE, LIGHT-EMITTING APPARATUS, ELECTRONIC DEVICE, AND LIGHTING DEVICE

Abstract

A light-emitting device with high emission efficiency is provided. A light-emitting device with a high blue index (BI) is provided. A light-emitting device with low power consumption is provided. A light-emitting device including a first electrode and a second electrode which are a reflective electrode and a semi-transmissive and semi-reflective electrode, and an EL layer sandwiched between the first electrode and the second electrode, where the EL layer contains an emission center substance, where when the emission center substance in the EL layer includes only one kind of substance, photon energy of a peak wavelength of light emitted from the light-emitting device is designed from an average value of photon energy of light emitted by the emission center substance in a solution state and emission edge energy on a short wavelength side of an emission spectrum of the emission center substance in the solution state.

Inventors: SEO; Satoshi (Sagamihara, JP), SASAKI; Toshiki (Isehara, JP), YAMAOKA; Ryohei (Atsugi, JP), NOMURA; Shiho (Atsugi, JP)

Applicant: SEMICONDUCTOR ENERGY LABORATORY CO., LTD. (ATSUGI-SHI, JP)

Family ID: 68539862

Assignee: SEMICONDUCTOR ENERGY LABORATORY CO., LTD. (ATSUGI-SHI, JP)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/586,676, filed Feb. 26, 2024, now allowed, which is incorporated by reference and is a continuation of U.S. application Ser. No. 17/055,933, filed Nov. 16, 2020, now U.S. Pat. No. 11,917,840, which is incorporated by reference and is a U.S. National Phase Application under 35 U.S.C. § 371 of International Application PCT/IB2019/053851, filed on May 10, 2019, which is incorporated by reference and claims the benefit of a foreign priority application filed in Japan on May 18, 2018, as Application No. 2018-096504.

TECHNICAL FIELD

[0002] One embodiment of the present invention relates to a light-emitting device, a light-emitting apparatus, an electronic device, and a lighting device. Note that one embodiment of the present invention is not limited to the above technical field. The technical field of one embodiment of the invention disclosed in this specification and the like relates to an object, a method, or a manufacturing method. One embodiment of the present invention relates to a process, a machine, manufacture, or a composition of matter. Specifically, examples of the technical field of one embodiment of the present invention disclosed in this specification include a semiconductor device, a display device, a liquid crystal display device, a light-emitting apparatus, a lighting device, a power storage device, a memory device, an imaging device, a driving method thereof, and a manufacturing method thereof.

BACKGROUND ART

[0003] Light-emitting devices (organic EL element) that use organic compounds and utilize electroluminescence (EL) have been put into practical use. The basic structure of such light-emitting devices is that an organic compound layer (EL layer) containing a light-emitting material is sandwiched between a pair of electrodes. Carriers are injected by application of voltage to this device, and recombination energy of the carriers is used, whereby light emission can be obtained from the light-emitting material.

[0004] Such light-emitting devices are of self-light-emitting type, and have advantages over liquid crystal, such as high visibility and no need for backlight when used for pixels of a display; accordingly, the light-emitting devices are suitable as flat panel display elements. Displays using such light-emitting devices are also highly advantageous in that they can be fabricated thin and lightweight. Moreover, an extremely fast response speed is also a feature.

[0005] Since light-emitting layers of such light-emitting devices can be successively formed two-

dimensionally, planar light emission can be obtained. This feature is difficult to obtain with point light sources typified by incandescent lamps and LEDs or linear light sources typified by fluorescent lamps. Thus, light-emitting devices are also of great utility value as planar light sources, which can be applied to lighting and the like.

[0006] Displays or lighting devices including light-emitting devices can be suitably used for a variety of electronic devices as described above, and research and development of light-emitting devices have progressed for better characteristics.

[0007] One of the important characteristics of light-emitting devices is emission efficiency typified by external quantum efficiency. The light-emitting device with favorable emission efficiency enables power consumption to be reduced. Thus, in view of world's energy situation becoming severer year by year, the higher the efficiency, the better.

REFERENCE

Patent Document

[0008] [Patent Document 1] Japanese Published Patent Application No. 2015-187982

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

[0009] An object of one embodiment of the present invention is to provide a novel light-emitting device. Alternatively, an object of another embodiment of the present invention is to provide a light-emitting device with high emission efficiency. Alternatively, an object of another embodiment of the present invention is to provide a light-emitting device with a high blue index (BI). An object of another embodiment of the present invention is to provide a light-emitting device with low power consumption.

[0010] Alternatively, an object of another embodiment of the present invention is to provide a light-emitting device, a light-emitting apparatus, an electronic device, a lighting device, and an electronic apparatus, each of which has low power consumption.

[0011] Note that the description of these objects does not preclude the existence of other objects. One embodiment of the present invention does not need to achieve all the objects. Other objects can be derived from the descriptions of the specification, the drawings, and the claims.

Means for Solving the Problems

[0012] One embodiment of the present invention is a light-emitting device including a first electrode, a second electrode, and an EL layer sandwiched between the first electrode and the second electrode, where one of the first electrode and the second electrode is a reflective electrode, and the other is a semi-transmissive and semi-reflective electrode, where the EL layer includes an emission center substance, where when an average value of photon energy of light emitted by the emission center substance in a solution state is denoted by $E_{\text{sub.ave}}$ [eV] and emission edge energy on a short wavelength side of an emission spectrum of the emission center substance in the solution state is denoted by $E_{\text{sub.edge}}$ [eV], photon energy $E_{\text{sub.em}}$ [eV] of a peak wavelength of light emitted from the light-emitting device is expressed by the following formula (1).

$$[00001] E_{\text{ave}} \leq E_{\text{em}} \leq 0.95E_{\text{edge}} \quad (1)$$

[0013] Alternatively, another embodiment of the present invention is a light-emitting device with the above structure, where the emission center substance exhibits blue light.

[0014] Alternatively, another embodiment of the present invention is a light-emitting device including a first electrode, a second electrode, an EL layer sandwiched between the first electrode and the second electrode, where one of the first electrode and the second electrode is a reflective electrode, and the other is a semi-transmissive and semi-reflective electrode, where the EL layer includes an emission center substance exhibiting blue emission, where when an average value of photon energy of light emitted by the emission center substance in a solution state is denoted by $E_{\text{sub.ave}}$ [eV] and emission edge energy on a short wavelength side of an emission spectrum of the emission center substance in the solution state is denoted by $E_{\text{sub.edge}}$ [eV], photon energy

E.sub.em [eV] of a peak wavelength of light emitted from the light-emitting device is expressed by the following formula (2).

$$[00002] \ E_{ave} + 0.02 \leq E_{em} \leq 0.95E_{edge} \quad (2)$$

[0015] Alternatively, another embodiment of the present invention is the light-emitting device with any of the above structures, where the E.sub.em is greater than or equal to 2.6 eV and less than or equal to 2.9 eV.

[0016] Alternatively, another embodiment of the present invention is the light-emitting device with any of the above structures, where an organic compound layer with a molecular weight greater than or equal to 300 and less than or equal to 1200 is formed on a surface of the semi-transmissive and semi-reflective electrode, which is an opposite surface facing the reflective electrode.

[0017] Alternatively, another embodiment of the present invention is the light-emitting device with any of the above structures, where a solvent in the solution state has a dielectric constant greater than or equal to 1 and less than or equal to 10, at room temperature.

[0018] Alternatively, another embodiment of the present invention is the light-emitting device with any of the above structures, where a solvent in the solution state is toluene or chloroform.

[0019] Alternatively, another embodiment of the present invention is the light-emitting device with any of the above structures, where the emission center substance in the EL layer includes only one kind of substance.

[0020] Alternatively, another embodiment of the present invention is a light-emitting apparatus including the light-emitting device described in any of the above and a transistor or a substrate.

[0021] Alternatively, another embodiment of the present invention is an electronic device including the above light-emitting apparatus, and a sensor, an operation button, a speaker, or a microphone.

[0022] Alternatively, another embodiment of the present invention is a lighting device including the light-emitting apparatus and a housing.

[0023] Alternatively, another embodiment of the present invention is an electronic device including an organic compound described in any of the above.

[0024] Note that the light-emitting apparatus in this specification includes an image display device using a light-emitting device. A module in which a connector such as an anisotropic conductive film or a TCP (Tape Carrier Package) is connected to a light-emitting device, a module in which a printed wiring board is provided on the tip of a TCP, and a module in which an IC (integrated circuit) is directly mounted on a light-emitting device by a COG (Chip On Glass) method include a light-emitting apparatus in some cases. Furthermore, in some cases, a lighting device or the like includes the light-emitting device.

Effect of the Invention

[0025] According to one embodiment of the present invention, a novel light-emitting device can be provided. According to another embodiment of the present invention, a light-emitting device having high emission efficiency can be provided. According to another embodiment of the present invention, a light-emitting device with high external quantum efficiency can be provided.

According to another embodiment of the present invention, a light-emitting device with a high blue index can be provided. According to another embodiment of the present invention, a light-emitting device with low power consumption can be provided.

[0026] According to another embodiment of the present invention, a light-emitting device, a light-emitting apparatus, an electronic device, a lighting device, and an electronic apparatus each having low power consumption can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIGS. 1(A) and 1(B) are diagrams showing a calculation method of an average value of

photon energy (E.sub.ave) of light emitted by an emission center substance in a solution state.

[0028] FIG. 2 is a diagram showing a calculation method of an emission edge energy (E.sub.edge) on a short wavelength side of an emission spectrum.

[0029] FIG. 3 is a diagram showing a relation of energy at an EL emission peak (E.sub.em), external quantum efficiency (EQE), and a blue index (BI).

[0030] FIGS. 4(A) and 4(B) are each a conceptual diagram of a light-emitting device.

[0031] FIGS. 5(A) and 5(B) are conceptual diagrams of an active matrix light-emitting apparatus.

[0032] FIGS. 6(A) and 6(B) are diagrams illustrating a lighting device.

[0033] FIGS. 7(A), 7(B1), 7(B2), and 7(C) are diagrams each illustrating an electronic device.

[0034] FIGS. 8(A) to 8(C) are diagrams each illustrating a light source device.

[0035] FIG. 9 is a diagram illustrating a lighting device.

[0036] FIG. 10 is a diagram illustrating a lighting device.

[0037] FIG. 11 is a diagram illustrating an in-vehicle display device and a lighting device.

[0038] FIGS. 12(A) and 12(B) are diagrams illustrating an electronic device.

[0039] FIGS. 13(A) to 13(C) are diagrams illustrating an electronic device.

[0040] FIG. 14 shows an emission spectrum of N,N'-diphenyl-N,N'-bis[3-(9-phenyl-9H-fluoren-9-yl)phenyl]pyrene-1,6-diamine (abbreviation: 1,6mFLPAPrn) in a toluene solution.

[0041] FIG. 15 is a diagram showing a calculation method of an emission edge on a short wavelength side of a PL spectrum of 1,6mFLPAPrn in a toluene solution.

[0042] FIG. 16 is a graph showing a relation of external quantum efficiency (EQE) and a blue index (BI) with respect to energy at emission peak (E.sub.em) in a light-emitting device 1-1 to a light-emitting device 1-8.

[0043] FIG. 17 is a diagram showing an emission spectrum of N,N'-bis(3-methylphenyl)-N,N'-bis[3-(9-phenyl-9H-fluoren-9-yl)phenyl]pyrene-1,6-diamine (abbreviation: 1.6 mMemFLPAPrn) in a toluene solution.

[0044] FIG. 18 is a diagram showing a calculation method of an emission edge on a short wavelength side of a PL spectrum of 1.6 mMemFLPAPrn in a toluene solution.

[0045] FIG. 19 is a graph showing a relation of external quantum efficiency (EQE) and a blue index (BI) with respect to energy at emission peak (E.sub.em) in a light-emitting device 2-1 to a light-emitting device 2-8.

[0046] FIG. 20 is a diagram showing an emission spectrum of N,N'-(pyrene-1,6-diyl)bis[(6,N-diphenylbenzo[b]naphtho[1,2-d]furan)-8-amine](abbreviation: 1,6BnfAPrn-03) in a toluene solution.

[0047] FIG. 21 is a diagram showing a calculation method of an emission edge on a short wavelength side of a PL spectrum of 1,6BnfAPrn-03 in a toluene solution.

[0048] FIG. 22 is a graph showing a relation of external quantum efficiency (EQE) and a blue index (BI) with respect to energy at emission peak (E.sub.em) in a light-emitting device 3-1 to a light-emitting device 3-8.

[0049] FIG. 23 is a diagram showing an emission spectrum of N,N'-(pyrene-1,6-diyl)bis[(6,N-diphenylbenzo[b]naphtho[1,2-d]furan)-8-amine](abbreviation: 1,6BnfAPrn-03) in a toluene solution.

[0050] FIG. 24 is a diagram showing a calculation method of an emission edge on a short wavelength side of a PL spectrum of 1,6BnfAPrn-03 in a toluene solution.

[0051] FIG. 25 is a graph showing a relation of external quantum efficiency (EQE) and a blue index (BI) with respect to energy at emission peak (E.sub.em) in a light-emitting device 4-1 to a light-emitting device 4-4.

[0052] FIG. 26 is a diagram showing an emission spectrum of N,N'-(pyrene-1,6-diyl)bis(N-phenyl-6-cyclohexylbenzo[b]naphtho[1,2-d]furan-8-amine) (abbreviation: 1,6chBnfAPrn) in a toluene solution.

[0053] FIG. 27 is a diagram showing a calculation method of an emission edge on a short

wavelength side of a PL spectrum of 1,6chBnfAPrn in a toluene solution.

[0054] FIG. **28** is a graph showing a relation of external quantum efficiency (EQE) and a blue index (BI) with respect to energy at emission peak (E.sub.em) in a light-emitting device **5-1** to a light-emitting device **5-8**.

[0055] FIG. **29** is a diagram showing an emission spectrum of 3,10-bis[N-(9-phenyl-9H-carbazol-2-yl)-N-phenylamino]naphtho[2,3-b; 6,7-b']bisbenzofuran (abbreviation: 3,10PCA2N bf(IV)-02) in a toluene solution.

[0056] FIG. **30** is a diagram showing a calculation method of an emission edge on a short wavelength side of a PL spectrum of 3,10PCA 2N bf(IV)-02 in a toluene solution.

[0057] FIG. **31** is a graph showing a relation of external quantum efficiency (EQE) and a blue index (BI) with respect to energy at emission peak (E.sub.em) in a light-emitting device **6-1** to a light-emitting device **6-8**.

[0058] FIG. **32** is a diagram showing an emission spectrum of 3,10-bis[N-(dibenzofuran-3-yl)-N-phenylamino]naphtho[2,3-b; 6,7-b']bisbenzofuran (abbreviation: 3,10FrA2Nbf(IV)-02) in a toluene solution.

[0059] FIG. **33** is a diagram showing a calculation method of an emission edge on a short wavelength side of a PL spectrum of 3,10FrA2Nbf(IV)-02 in a toluene solution.

[0060] FIG. **34** is a graph showing a relation of external quantum efficiency (EQE) and a blue index (BI) with respect to energy at emission peak (E.sub.em) in a light-emitting device **7-1** to a light-emitting device **7-8**.

MODE FOR CARRYING OUT THE INVENTION

[0061] Embodiments of the present invention are described in detail below with reference to drawings. Note that the present invention is not limited to the following description, and it will be readily appreciated by those skilled in the art that modes and details of the present invention can be modified in various ways without departing from the spirit and scope of the present invention. Thus, the present invention should not be construed as being limited to the descriptions in the following embodiments.

Embodiment 1

[0062] A variety of light-emitting devices exists as devices emitting light. Whereas an LED uses an inorganic compound as an emission center, an OLED, what is called organic EL element, is a light-emitting device using an organic compound as an emission center substance. The organic compound is known to exhibit an emission spectrum with a broader shape, due to influence of a vibration level or the like, than the case of the inorganic compound.

[0063] To display images with high color reproducibility with a display element such as a display, it is necessary to use a display element emitting light with favorable color purity. Thus, in the case where an organic EL element with a broad emission spectrum as described above and relatively low color purity is used for a display element for displaying images, a color filter, a microcavity structure, or the like is used, in some cases, to improve the color purity.

[0064] As compared with using a color filter simply blocking light emission, using the microcavity structure that enables amplification of light with an aimed wavelength less loses emission and is advantageous in efficiency. Furthermore, in combination of the light-emitting device having a microcavity structure with an appropriate optical path length and the color filter, the amount of light whose wavelength is blocked can be reduced, so that a light-emitting device having high efficiency and emitting light with high color purity can be obtained. As described above, employing the microcavity structure is suitable for an improvement in color purity and maintaining emission efficiency.

[0065] The organic EL element is a light-emitting device using an organic compound as an emission center substance as mentioned above. A spectrum of light emitted by the organic compound has a wavelength and shape derived from a structure of the organic compound, and is specific to the substance. Thus, the wavelength range where light emission can be efficiently

obtained differs depending on an emission center substance used in the light-emitting device.

[0066] The present inventors found that in light-emitting devices with a microcavity structure, a light-emitting device satisfying the following formula (1) is a light-emitting device with favorable emission efficiency (quantum efficiency calculated from front luminance).

$$[00003] E_{\text{ave}} \leq E_{\text{em}} \leq 0.95E_{\text{edge}} \quad (1)$$

[0067] Note that in the above formula, E.sub.ave is an average value of photon energy of light emitted by an emission center substance, which is used in the light-emitting device, in a solution state; E.sub.em is photon energy of a peak wavelength of light emitted from the light-emitting device; and E.sub.edge is emission edge energy on a short wavelength side of an emission spectrum of the emission center substance in the solution state. The unit of all of them is [eV].

<Method for Calculating Average Value of Photon Energy (E.SUB.ave.) of Light Emitted by Emission Center Substance in Solution State>

[0068] A method for calculating E.sub.ave is described with reference to FIG. 1(A). FIG. 1(A) shows a normalized emission spectrum of an organic compound emitting blue light in a toluene solution. The vertical axis is proportional to a photon count. When the emission spectrum is represented as a function of F(λ) (λ is wavelength), the number of photons N.sub.p(λ) per unit time of each wavelength is N.sub.p(λ)=F.sub.0F (F.sub.0 is a proportionality constant). Thus, a total number of photons per unit time of the organic compound in the toluene solution can be expressed by the following formula (1). Note that, $\int F(\lambda)d\lambda$ in the following formula (1) corresponds to an integral value of the emission spectrum of FIG. 1(A).

$$[00004] N_p = \int N_p(\lambda) d\lambda = F_0 \int F(\lambda) d\lambda \quad (I)$$

[0069] Meanwhile an energy $\phi_{\text{sub.p}}(\lambda)$ per unit time of photons in a certain wavelength can be expressed by the following formula (II). Note that in the following formula (II), c denotes light speed and h denotes the Planck constant.

$$[00005] \phi_p(\lambda) = N_p(\lambda) \cdot \text{Math. ch} / \lambda = F_0 F(\lambda) \cdot \text{Math. ch} / \lambda \quad (II)$$

[0070] Therefore, the total energy in all wavelength regions per unit time for light emission of the organic compound in the toluene solution can be expressed by the following formula (III).

$$[00006] \phi_p = \int \phi_p(\lambda) d\lambda = F_0 \text{ch} \int F(\lambda) / \lambda d\lambda \quad (III)$$

[0071] The average photon energy (E.sub.ave) of light emission is a value obtained by dividing the total energy expressed by the above formula (III) by the total number of photons expressed by the above formula (1), and thus can be expressed by the following formula (IV).

[0072] [Equation 1]

$$[00007] E_{\text{ave}} = \frac{\phi_p}{N_p} = \text{ch} \cdot \frac{\int F(\lambda) / \lambda d\lambda}{\int F(\lambda) d\lambda} \quad (IV)$$

[0073] The average photon energy (E.sub.ave) of light emission of the organic compound in the toluene solution, whose emission spectrum is shown in FIG. 1(A), can be calculated from the above formula (IV) to be 2.65 eV, i.e., 468 nm (FIG. 1(B)). Note that the significant digit is three.

[0074] Here, depending on used calculation tools, in some cases, the vertical axis of the PL spectrum is not photon number N.sub.p(λ) but energy $\phi_{\text{sub.p}}(\lambda)$ itself (in the case of a spectra radiance meter, for example) or normalized spectrum I(A) that is proportional to the energy $\phi_{\text{sub.p}}(\lambda)$. In such a case, the average photon energy E.sub.ave of light emission can be calculated as described below.

[0075] First, based on the above formulae (1) and (II), the total numbers of photons N.sub.p per unit time can be expressed by the following formula (V).

$$[00008] N_p = \int N_p(\lambda) d\lambda = (1 / \text{ch}) \int \phi_p(\lambda) d\lambda \quad (V)$$

[0076] Thus, in the case where the vertical axis of the PL spectrum is energy $\phi_{\text{sub.p}}(\lambda)$ itself, the average photon energy (E.sub.ave) of light emission can be calculated from the following formula

(VI) based on the formulae (III) to (V).

[00009] [Equation2]
$$E_{\text{ave}} = \frac{\phi_p}{N_p} = \text{ch. Math.} \frac{\int \phi_p(\lambda) d\lambda}{\int \phi_p(\lambda) d\lambda} \quad (\text{VI})$$

[0077] Alternatively, in the case where the normalized spectrum $I(\lambda)$ that is proportional to the energy is used, calculation can be made as the following formula (VII).

[00010] [Equation3]
$$E_{\text{ave}} = \frac{\phi_p}{N_p} = \text{ch. Math.} \frac{\int I(\lambda) d\lambda}{\int I(\lambda) d\lambda} \quad (\text{VII})$$

[0078] As described above, the average photon energy ($E_{\text{sub.ave}}$) of light emission can be calculated from any one of the above formulae (IV), (VI), and (VII), depending on a toll for measuring an emission spectrum of an organic compound in a toluene solution.

<Method for Calculating Emission Edge Energy ($E_{\text{sub.edge}}$) on Short Wavelength Side of Emission Spectrum>

[0079] A method for calculating $E_{\text{sub.edge}}$ is described. FIG. 2 shows an enlarged graph in the vicinity of emission edge on the short wavelength side of the emission center substance in FIG. 1. As shown in FIG. 2, $E_{\text{sub.edge}}$ can be determined by drawing a tangent to at around a half of the peak toward a tail on the short wavelength side of emission spectrum $F(\lambda)$ and calculating an intersection of the tangent and the x axis.

[0080] Since the intersection of the x axis and the tangent drawn to at around the half of the peak toward the tail on the short wavelength side of the emission spectrum shown in FIG. 2 is determined and found to be 432 nm, the emission edge energy ($E_{\text{sub.edge}}$) on the short wavelength side of the emission spectrum can be figured out as 2.87 eV.

[0081] Note that the PL spectrum of the emission center substance used for calculation of $E_{\text{sub.ave}}$ and $E_{\text{sub.edge}}$ is the spectrum of the emission center substance in a solution state. Each energy level is affected by the dielectric constant or the like. The dielectric constant of an organic compound included in the EL layer of the light-emitting device is approximately 3; therefore, measurement is conducted, at room temperature, in a solution where the dielectric constant is greater than or equal to 1 and less than or equal to 10, preferably greater than or equal to 2 and less than or equal to 5, whereby a fairly accurate value can be obtained. Specific examples of a solvent include hexane, benzene, toluene, diethyl ether, ethyl acetate, chloroform, chlorobenzene, and dichloromethane. In particular, toluene and chloroform each have a dielectric constant greater than or equal to 2 and less than or equal to 5, at room temperature, have high solubility and be used as a versatile solvent. Thus, when measurement is conducted in a solvent of toluene or chloroform, a difference between a value measured in the light-emitting device and a value measured in the solvent is within an experimental error range in many cases.

<Relation Between EL Emission Peak Energy ($E_{\text{sub.em}}$), External Quantum Efficiency (EQE), and Blue Index (BI) of Light-Emitting Device>

[0082] FIG. 3 shows an examined relation between $E_{\text{sub.em}}$ and EQE and an examined relation between $E_{\text{sub.em}}$ and BI of an organic EL element (top emission) with a microcavity structure using an emission substance used in FIG. 1 and FIG. 2 as an emission center. Note that EQE shown here is an uncorrected EQE calculated from front luminance on the assumption of Lambertian distribution. In addition, the EQE is one when the luminance is around 1000 cd/m.

[0083] Note that the blue index (BI) is a value obtained by further dividing current efficiency (cd/A) by chromaticity y , and is one of the indicators representing characteristics of blue light emission. As the chromaticity y is smaller, the color purity of blue light emission tends to be higher. With high color purity for blue light emission, a wide range of blue can be expressed even with a small number of luminance components; thus, using blue light emission with high color purity reduces the luminance needed for expressing blue, leading to lower power consumption. Thus, BI that is based on chromaticity y , which is one of the indicators of color purity of blue, is suitably used as a mean for showing efficiency of blue light emission. The light-emitting device with higher BI can be regarded as a blue light-emitting device having more favorable efficiency for a display.

[0084] The EL emission peak energy (E.sub.em) shown in the horizontal axis of FIG. 3 can be controlled by changing an optical path length of the organic EL element and changing a wavelength amplified with the microcavity effect. As a result of plotting EQE with respect to the E.sub.em, favorable efficiency is shown in a range from 2.65 eV to 2.73 eV (range of A in the drawing), and EQE of the light-emitting device exhibiting E.sub.em beyond the range greatly decreases. In addition, BI also exhibits favorable values in a similar range. As calculated in the above, 2.65 eV corresponds to the average photon energy (E.sub.ave) of light emission of the emission center substance which is in the toluene solution and used in the light-emitting device.

[0085] The upper limit of the range A in the drawing, 2.73 eV, corresponds to 0.95 times the value of 2.87 eV calculated above, the emission edge energy (E.sub.edge) on the short wavelength side of a PL emission spectrum of the emission center substance which is in the toluene solution and used for the above organic EL element.

[0086] In the above manner, in the organic EL element with a microcavity structure, the cavity length (optical path length) in the microcavity structure is controlled to satisfy a relation shown in the following formula (1), whereby a light-emitting device with favorable EQE can be obtained. Note that, in this range, BI also exhibits favorable values.

$$[00011] E_{\text{ave}} \leq E_{\text{em}} \leq 0.95E_{\text{edge}} \quad (1)$$

(Note that E.sub.ave represents the average photon energy (unit [eV]) of light emitted by an emission center substance in a toluene solution, which is used for a light-emitting device, E.sub.edge represents the emission edge energy (unit [eV]) on a short wavelength side of a PL emission spectrum of the emission center substance in the toluene solution, and E.sub.em represents the EL emission peak energy (unit [eV]).)

[0087] In the case of an organic EL element emitting blue light, it is appropriate to focus on BI. The present inventors found that BI and EQE have different peak values of E.sub.em. As shown in FIG. 3, BI has a maximum value when E.sub.em is positioned at a relatively higher region than the case of EQE; thus, the cavity length (optical length) of the microcavity structure is preferably controlled to satisfy the following formula (2).

$$[00012] E_{\text{ave}} + 0.02 \leq E_{\text{em}} \leq 0.95E_{\text{edge}} \quad (2)$$

[0088] Note that the organic EL element of one embodiment of the present invention preferably has E.sub.em greater than or equal to 2.6 eV and less than or equal to 2.9 eV.

[0089] Furthermore, an organic EL element of one embodiment of the present invention is a light-emitting device with a microcavity structure. The organic EL element emits light by making current flow to a structure where an EL layer containing an organic compound is sandwiched between a pair of electrodes. Meanwhile, the microcavity structure is such a structure that one of a pair of electrodes is a reflective electrode and the other is a semi-transmissive and semi-reflective electrode, which enables repetition of reflection and amplification of light with a wavelength corresponding to a distance between the electrodes (also referred to as cavity length or optical path length). The cavity length can be changed by adjusting the thickness of an EL layer or the electrode. In the case where the cavity length is adjusted with the electrode, a transparent electrode such as ITO can be used. In the case where the EL layer is adjusted, the thickness of a carrier transport layer or a carrier injection layer is adjusted, so that the optical path length can be controlled.

[0090] As the light emission direction of the organic EL element, either a top-emission structure or a bottom-emission structure may be employed. In the top-emission light-emitting device, an organic compound layer with a molecular weight greater than or equal to 300 and less than or equal to 1200 is preferably formed on a surface of a semi-transmissive and semi-reflective electrode, which is an opposite surface facing the reflective electrode, so as to conduct light extraction more efficiently.

[0091] Note that in the light-emitting device of one embodiment of the present invention, one kind

of emission center substance is preferable for an emission center substance in one light-emitting device.

[0092] A light-emitting device of one embodiment of the present invention with the above structure can be a light-emitting device with favorable emission efficiency.

[0093] Next, an example of a light-emitting device which is one embodiment of the present invention will be described in detail below with reference to FIG. 4(A).

[0094] The light-emitting device in this embodiment includes a pair of electrodes of a first electrode **101** and a second electrode **102** and an EL layer **103** provided between the first electrode **101** and the second electrode **102**. In FIG. 4(A), the electrode provided on the formation substrate side is referred to as the first electrode **101**.

[0095] The light-emitting device of one embodiment of the present invention is a light-emitting device having a microcavity structure. The light-emitting device with a microcavity structure can be obtained with use of a reflective electrode and a semi-transmissive and semi-reflective electrode as a pair of electrodes. The reflective electrode and the semi-transmissive and semi-reflective electrode correspond to the first electrode **101** and the second electrode **102** described above. The light-emitting device includes at least an EL layer between the reflective electrode and the semi-transmissive and semi-reflective electrode, which includes at least a light-emitting layer serving as a light-emitting region.

[0096] In the light-emitting device with a microcavity structure, light emitted in all directions from the light-emitting layer included in the EL layer is reflected and resonated by the reflective electrode and the semi-transmissive and semi-reflective electrode.

[0097] Note that the reflective electrode has a visible light reflectivity of 40% to 100%, preferably 70% to 100%, and a resistivity of $1 \times 10^{-2} \Omega\text{cm}$ or lower. Examples of a material forming the reflective electrode include aluminum (Al) or an alloy containing Al. Examples of the alloy containing Al include an alloy containing Al and L (L represents one or more of titanium (Ti), neodymium (Nd), nickel (Ni), and lanthanum (La)), such as an alloy containing Al and Ti and an alloy containing Al, Ni, and La. Aluminum has low resistance and high light reflectivity. Aluminum is included in earth's crust in large amount and is inexpensive; therefore, it is possible to reduce costs for manufacturing a light-emitting device with aluminum. Furthermore, silver (Ag), an alloy containing Ag and N (N represents one or more of yttrium (Y), Nd, magnesium (Mg), ytterbium (Yb), Al, Ti, gallium (Ga), zinc (Zn), indium (In), tungsten (W), manganese (Mn), tin (Sn), iron (Fe), Ni, copper (Cu), palladium (Pd), iridium (Ir), and gold (Au)), or the like may be used. Examples of the alloy containing silver include an alloy containing silver, palladium, and copper, an alloy containing silver and copper, an alloy containing silver and magnesium, an alloy containing silver and nickel, an alloy containing silver and gold, and an alloy containing silver and ytterbium. Besides, a transition metal such as tungsten, chromium (Cr), molybdenum (Mo), copper, or titanium can be used. Note that a transparent electrode layer is formed using a light-transmitting conductive material between the reflective electrode so that the first electrode **101** can have a two-layer structure of the reflective electrode and the transparent electrode. By this, an optical length of the microcavity structure (microcavity length) can be adjusted. Examples of the light-transmitting conductive material include a metal oxide such as indium tin oxide (hereinafter, referred to as ITO), indium tin oxide containing silicon or silicon oxide (abbreviation: ITSO), indium oxide-zinc oxide (Indium Zinc Oxide), indium oxide-tin oxide containing titanium, indium titanium oxide, or indium oxide containing tungsten oxide and zinc oxide can be used. In FIG. 4(A), first electrode **101** is formed with a reflective electrode **101-1** and a transparent electrode **101-2**.

[0098] In addition, the semi-transmissive and semi-reflective electrode has a visible light reflectivity of 20% to 80%, preferably 40% to 70%, and a resistivity of $1 \times 10^{-2} \Omega\text{cm}$ or lower. For example, one or more kinds of conductive metals and alloys, conductive compounds, and the like can be used as the semi-transmissive and semi-reflective electrode. Specifically, a metal oxide such as indium tin oxide (hereinafter, referred to as ITO), indium tin oxide containing silicon or

silicon oxide (abbreviation: ITO), indium oxide-zinc oxide (Indium Zinc Oxide), indium oxide-tin oxide containing titanium, indium titanium oxide, or indium oxide containing tungsten oxide and zinc oxide can be used. A metal thin film having a thickness that allows transmission of light (preferably, a thickness greater than or equal to 1 nm and less than or equal to 30 nm) can also be used. As the metal, Ag, an alloy of Ag and Al, an alloy of Ag and Mg, an alloy of Ag and Au, an alloy of Ag and Yb, or the like can be used.

[0099] The reflective electrode can be either one of the first electrode **101** and the second electrode **102** and the semi-transmissive and semi-reflective electrode can be the other. FIG. 4(A) shows a case where the first electrode **101** is on the formation substrate side as described above; thus, the light-emitting device has a top emission structure when the reflective electrode is the first electrode, and the light-emitting device is a bottom emission device when the reflective electrode is the second electrode **102**. Note that although the first electrode **101** and the second electrode **102** may serve as either an anode or a cathode, FIG. 4(A) shows a case where the first electrode **101** is an anode.

[0100] Note that when the light-emitting device has a top-emission structure, the light extraction efficiency can be improved by providing an organic cap layer **104** on a surface of the second electrode **102** which is opposite to a surface in contact with the EL layer **103**. When the organic cap layer **104** is provided in contact with the electrode **102** in the light-emitting device, the difference in refractive index at the interface between the electrode **102** and the air can be reduced, resulting in improvement in light extraction efficiency. The thickness is preferably greater than or equal to 5 nm and less than or equal to 120 nm. Further preferably, the thickness is greater than or equal to 30 nm and less than or equal to 90 nm. An organic compound layer including a substance with a molecular weight greater than or equal to 300 and less than or equal to 1200 is preferably used as the organic cap layer **104**. Furthermore, the organic cap layer is preferably formed using a conductive organic material. In the structure, the second electrode **102** is the semi-transmissive and semi-reflective electrode, which needs a thin thickness so as to ensure a certain light-transmitting property, and conductivity might be decreased in some cases. With use of a conductive material for organic cap layer **104**, while the light extraction efficiency is improved, the conductivity can be ensured and the manufacturing yield of the light-emitting device can be improved. Note that an organic compound that is less likely to absorb light in a visible light region can be favorably used. For the organic cap layer, the organic compound used for the EL layer **103** can also be used for the organic cap layer **104**. In that case, the organic cap layer **104** can be formed with a deposition apparatus or a deposition chamber for forming the EL layer **103**, so that the organic cap layer **104** can be easily formed.

[0101] In the light-emitting device, by changing the thickness of the transparent electrode provided in contact with the above-described reflective electrode and the thicknesses of carrier-transport layers, such as a hole-injection layer and a hole-transport layer, the optical path length (cavity length) between the reflective electrode and the semi-transmissive and semi-reflective electrode can be changed. Thus, light with a wavelength that is resonated between the reflective electrode and the semi-transmissive and semi-reflective electrode can be intensified while light with a wavelength that is not resonated therebetween can be attenuated. FIG. 4(A) illustrates an example in which the optical path length is adjusted by the transparent electrode **101-2**, which is a part of the first electrode **101**; however, the optical path length may be adjusted by a hole-injection layer **111** as shown in FIG. 4(B) or by a hole-transport layer **112**, or a combination of two or more of them may be used.

[0102] Note that in light emission, light that is reflected back by the reflective electrode (first reflected light) considerably interferes with light that directly enters the semi-transmissive and semi-reflective electrode from the light-emitting layer (first incident light); therefore, the optical path length between the reflective electrode and the light-emitting layer is preferably adjusted to $(2n-1)\lambda/4$ (n is a natural number of 1 or larger and λ is a wavelength of light emission to be

amplified). By adjusting the optical path length, the phases of the first reflected light and the first incident light can be aligned with each other and the light emitted from the light-emitting layer can be further amplified.

[0103] With the microcavity structure, emission intensity with a specific wavelength in the front direction can be increased, whereby power consumption can be reduced.

[0104] Although the EL layer **103** preferably has a stacked structure, there is no particular limitation on the stacked structure, and various layer structures such as a hole-injection layer, a hole-transport layer, an electron-transport layer, an electron-injection layer, a carrier-blocking layer, an exciton-blocking layer, and a charge-generation layer can be employed. In this embodiment, two kinds of stacked-layer structures are described: the structure including the electron-transport layer **114** and the electron-injection layer **115** in addition to the hole-injection layer **111**, the hole-transport layer **112**, and the light-emitting layer **113** as illustrated in FIG. 4(A); and the structure including the electron-transport layer **114**, the electron-injection layer **115**, and a charge-generation layer **116** in addition to the hole-injection layer **111**, the hole-transport layer **112**, and the light-emitting layer **113** as illustrated in FIG. 4(B). Materials forming the layers are specifically described below.

[0105] The hole-injection layer **111** is a layer containing a substance having an acceptor property. Either an organic compound or an inorganic compound can be used as the substance having an acceptor property.

[0106] As the substance having an acceptor property, a compound including an electron-withdrawing group (a halogen group or a cyano group) can be used; for example, 7,7,8,8-tetracyano-2,3,5,6-tetrafluoroquinodimethane (abbreviation: F4-TCNQ), 3,6-difluoro-2,5,7,7,8,8-hexacyanoquinodimethane, chloranil, 2,3,6,7,10,11-hexacyano-1,4,5,8,9,12-hexaazatriphenylene (abbreviation: HAT-CN), or 1,3,4,5,7,8-hexafluorotetracyano-naphthoquinodimethane (abbreviation: F6-TCNNQ) can be used. The organic compound having an acceptor property is preferably a compound in which electron-withdrawing groups are bonded to a condensed aromatic ring having a plurality of heteroatoms, like HAT-CN, because it is thermally stable. A [3]radialene derivative including an electron-withdrawing group (in particular, a cyano group or a halogen group such as a fluoro group) has a very high electron-accepting property and thus is preferable; specific examples include α,α',α'' -1,2,3-cyclopropanetriylidenetris[4-cyano-2,3,5,6-tetrafluorobenzeneacetonitrile], α,α',α'' -1,2,3-cyclopropanetriylidenetris[2,6-dichloro-3,5-difluoro-4-(trifluoromethyl)benzeneacetonitrile], and α,α',α'' -1,2,3-cyclopropanetriylidenetris[2,3,4,5,6-pentafluorobenzeneacetonitrile]. As the substance having an acceptor property, molybdenum oxide, vanadium oxide, ruthenium oxide, tungsten oxide, manganese oxide, or the like can be used, other than the above-described organic compounds. Alternatively, the hole-injection layer **111** can be formed using a phthalocyanine-based complex compound such as phthalocyanine (abbreviation: H.sub.2Pc) or copper phthalocyanine (CuPC), an aromatic amine compound such as 4,4'-bis[N-(4-diphenylaminophenyl)-N-phenylamino]biphenyl (abbreviation: DPAB) or N,N'-bis{4-[bis(3-methylphenyl)amino]phenyl}-N,N'-diphenyl-(1,1'-biphenyl)-4,4'-diamine (abbreviation: DNTPD), or a high molecular compound such as poly(3,4-ethylenedioxythiophene)/poly(styrenesulfonic acid) (PEDOT/PSS). The substance having an acceptor property can extract electrons from an adjacent hole-transport layer (or hole-transport material) when an electric field is applied.

[0107] Alternatively, a composite material in which a substance having a hole-transport property contains an acceptor substance can be used for the hole-injection layer **111**. By using a composite material in which a substance having a hole-transport property contains an acceptor substance, a material used to form an electrode can be selected regardless of its work function. In other words, besides a material having a high work function, a material having a low work function can also be used for the first electrode **101**. Examples of the acceptor substance include an organic compound having an acceptor property, such as 7,7,8,8-tetracyano-2,3,5,6-tetrafluoroquinodimethane (abbreviation: F4-TCNQ), chloranil, or 1,3,4,5,7,8-hexafluorotetracyano-naphthoquinodimethane

(abbreviation: F6-TCNNQ), and a transition metal oxide. In addition, oxides of metals belonging to Group 4 to Group 8 of the periodic table can be used. As the oxide of a metal belonging to Group 4 to Group 8 in the periodic table, vanadium oxide, niobium oxide, tantalum oxide, chromium oxide, molybdenum oxide, tungsten oxide, manganese oxide, rhenium oxide, or the like is preferably used because their electron-accepting property is high. Among these oxides, molybdenum oxide is particularly preferable because it is stable in the air, has a low hygroscopic property, and is easy to handle.

[0108] As the substance having a hole-transport property which is used for the composite material, a variety of organic compounds such as an aromatic amine compound, a carbazole derivative, an aromatic hydrocarbon, and a high molecular compound (an oligomer, a dendrimer, a polymer, or the like) can be used. Note that a substance having a hole mobility of 1×10^{-6} cm²/Vs or higher is preferably used as the substance having a hole-transport property which is used for the composite material. Organic compounds that can be used as the substance having a hole-transport property in the composite material are specifically given below.

[0109] Examples of the aromatic amine compounds that can be used for the composite material include N,N'-di(p-tolyl)-N,N'-diphenyl-p-phenylenediamine (abbreviation: DTDPPA), 4,4'-bis[N-(4-diphenylaminophenyl)-N-phenylamino]biphenyl (abbreviation: DPAB), N,N'-bis{4-[bis(3-methylphenyl)amino]phenyl}-N,N'-diphenyl-(1,1'-biphenyl)-4,4'-diamine (abbreviation: DNTPD), 1,3,5-tris[N-(4-diphenylaminophenyl)-N-phenylamino]benzene (abbreviation: DPA3B), and 1,1-bis-(4-bis(4-methyl-phenyl)-amino-phenyl)-cyclohexane (abbreviation: TAPC). Specific examples of the carbazole derivatives include 3-[N-(9-phenylcarbazol-3-yl)-N-phenylamino]-9-phenylcarbazole (abbreviation: PCzPCA1), 3,6-bis[N-(9-phenylcarbazol-3-yl)-N-phenylamino]-9-phenylcarbazole (abbreviation: PCzPCA2), 3-[N-(1-naphthyl)-N-(9-phenylcarbazol-3-yl)amino]-9-phenylcarbazole (abbreviation: PCzPCN1), 4,4'-di(N-carbazolyl)biphenyl (abbreviation: CBP), 1,3,5-tris[4-(N-carbazolyl)phenyl]benzene (abbreviation: TCPB), 9-[4-(10-phenylanthracene-9-yl)phenyl]-9H-carbazole (abbreviation: CzPA), and 1,4-bis[4-(N-carbazolyl)phenyl]-2,3,5,6-tetraphenylbenzene. Examples of the aromatic hydrocarbon include 2-tert-butyl-9,10-di(2-naphthyl)anthracene (abbreviation: t-BuDNA), 2-tert-butyl-9,10-di(1-naphthyl)anthracene, 9,10-bis(3,5-diphenylphenyl)anthracene (abbreviation: DPPA), 2-tert-butyl-9,10-bis(4-phenylphenyl)anthracene (abbreviation: t-BuDBA), 9,10-di(2-naphthyl)anthracene (abbreviation: DNA), 9,10-diphenylanthracene (abbreviation: DPAnth), 2-tert-butylanthracene (abbreviation: t-BuAnth), 9,10-bis(4-methyl-1-naphthyl)anthracene (abbreviation: DMNA), 2-tert-butyl-9,10-bis[2-(1-naphthyl)phenyl]anthracene, 9,10-bis[2-(1-naphthyl)phenyl]anthracene, 2,3,6,7-tetramethyl-9,10-di(1-naphthyl)anthracene, 2,3,6,7-tetramethyl-9,10-di(2-naphthyl)anthracene, 9,9'-bianthryl, 10,10'-diphenyl-9,9'-bianthryl, 10,10'-bis(2-phenylphenyl)-9,9'-bianthryl, 10,10'-bis[(2,3,4,5,6-pentaphenyl)phenyl]-9,9'-bianthryl, anthracene, tetracene, rubrene, perylene, and 2,5,8,11-tetra(tert-butyl)perylenene. Besides, pentacene, coronene, and the like can be used. A vinyl skeleton may be included. Examples of the aromatic hydrocarbon having a vinyl group include 4,4'-bis(2,2-diphenylvinyl)biphenyl (abbreviation: DPVBi) and 9,10-bis[4-(2,2-diphenylvinyl)phenyl]anthracene (abbreviation: DPV PA). Note that the organic compound of one embodiment of the present invention can also be used. In this case, F6-TCNNQ is preferably used as the acceptor substance.

[0110] It is also possible to use high molecular compounds such as poly(N-vinylcarbazole) (abbreviation: PVK), poly(4-vinyltriphenylamine) (abbreviation: PVTPA), poly[N-(4-{N'-[4-(4-diphenylamino)phenyl]phenyl}-N'-phenylamino)phenyl]methacrylamide (abbreviation: PTPDMA), and poly[N,N'-bis(4-butylphenyl)-N,N'-bis(phenyl)benzidine] (abbreviation: Poly-TPD).

[0111] The formation of the hole-injection layer **111** can improve the hole-injection property, whereby a light-emitting device having a low driving voltage can be obtained. The organic compound having an acceptor property is an easy-to-use material because evaporation is easy and its film can be easily formed.

[0112] Note that in the case where the hole-injection layer is formed using the above composite material, the layer containing the composite material is extremely appropriate for a layer adjusting the cavity length in the microcavity structure because the composite material has favorable conductivity and is less likely to cause a degradation of driving voltage even if the layer thickness is increased.

[0113] The hole-transport layer **112** contains a hole-transport material. The hole-transport material preferably has a hole mobility of 1×10^{-6} cm²/Vs or more. The hole-transport layer **112** preferably includes the organic compound of one embodiment of the present invention. The hole-transport layer **112** including the organic compound described in Embodiment 1 can be a layer with a low refractive index in the EL layer **103**, leading to higher external quantum efficiency of the light-emitting device.

[0114] As a material having a hole-transport property, a compound having an aromatic amine skeleton, such as 4,4'-bis[N-(1-naphthyl)-N-phenylamino]biphenyl (abbreviation: NPB), N,N'-bis(3-methylphenyl)-N,N'-diphenyl-[1,1'-biphenyl]-4,4'-diamine (abbreviation: TPD), 4,4'-bis[N-(spiro-9,9'-bifluoren-2-yl)-N-phenylamino]biphenyl (abbreviation: BSPB), 4-phenyl-4'-(9-phenylfluoren-9-yl)triphenylamine (abbreviation: BPAFLP), 4-phenyl-3'-(9-phenylfluoren-9-yl)triphenylamine (abbreviation: mBPAFLP), 4-phenyl-4'-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBA1BP), 4,4'-diphenyl-4'-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBBilBP), 4-(1-naphthyl)-4'-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBANB), 4,4'-di(1-naphthyl)-4'-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBN BB), 9,9-dimethyl-N-phenyl-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-fluoren-2-amine (abbreviation: PCBAF), or N-phenyl-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-spiro-9,9'-bifluoren-2-amine (abbreviation: PCBASF), a compound having a carbazole skeleton, such as 1,3-bis(N-carbazolyl)benzene (abbreviation: mCP), 4,4'-di(N-carbazolyl)biphenyl (abbreviation: CBP), 3,6-bis(3,5-diphenylphenyl)-9-phenylcarbazole (abbreviation: CzTP), or 3,3'-bis(9-phenyl-9H-carbazole) (abbreviation: PCCP), a compound having a thiophene skeleton, such as 4,4',4''-(benzene-1,3,5-triyl)tri(dibenzothiophene) (abbreviation: DBT3P-II), 2,8-diphenyl-4-[4-(9-phenyl-9H-fluoren-9-yl)phenyl]dibenzothiophene (abbreviation: DBTFLP-III), or 4-[4-(9-phenyl-9H-fluoren-9-yl)phenyl]-6-phenyldibenzothiophene (abbreviation: DBTFLP-IV), and a compound having a furan skeleton, such as 4,4',4''-(benzene-1,3,5-triyl)tri(dibenzofuran) (abbreviation: DBF3P-II) or 4-{3-[3-(9-phenyl-9H-fluoren-9-yl)phenyl]phenyl}dibenzofuran (abbreviation: mmDBFFLBi-II) can be given. Among the above, the compound having an aromatic amine skeleton and the compound having a carbazole skeleton are preferable because these have favorable reliability, have high hole-transport properties, and contribute to a reduction in driving voltage. Note that any of the substances given as examples of the material having a hole-transport property that is used in the composite material for the hole-injection layer **111** can also be suitably used as the material included in the hole-transport layer **112**.

[0115] The light-emitting layer **113** is a layer containing the host material and the light-emitting material. The light-emitting material may be fluorescent substances, phosphorescent substances, substances exhibiting thermally activated delayed fluorescence (TADF), or other light-emitting materials. Furthermore, it may be a single layer or be formed of a plurality of layers including different light-emitting materials.

[0116] Examples of a material that can be used as a fluorescent substance in the light-emitting layer **113** are as follows. Fluorescent substances other than those given below can also be used.

[0117] For example, 5,6-bis[4-(10-phenyl-9-anthryl)phenyl]-2,2'-bipyridine (abbreviation: PAP2BPy), 5,6-bis[4'-(10-phenyl-9-anthryl)biphenyl-4-yl]-2,2'-bipyridine (abbreviation: PAPP2BPy), N,N'-diphenyl-N,N'-bis[4-(9-phenyl-9H-fluoren-9-yl)phenyl]pyrene-1,6-diamine (abbreviation: 1,6FLPAPrn), N,N'-bis(3-methylphenyl)-N,N'-bis[3-(9-phenyl-9H-fluoren-9-yl)phenyl]-pyrene-1,6-diamine (abbreviation: 1.6 mMemFLPAPrn), N,N'-bis[4-(9H-carbazol-9-yl)phenyl]-N,N'-diphenylstilbene-4,4'-diamine (abbreviation: YGA2S), 4-(9H-carbazol-9-yl)-4'-

(10-phenyl-9-anthryl)triphenylamine (abbreviation: YGAPA), 4-(9H-carbazol-9-yl)-4'-(9,10-diphenyl-2-anthryl)triphenylamine (abbreviation: 2YGAPPA), N,9-diphenyl-N-[4-(10-phenyl-9-anthryl)phenyl]-9H-carbazol-3-amine (abbreviation: PCA PA), perylene, 2,5,8,11-tetra-tert-butyl perylene (abbreviation: TBP), 4-(10-phenyl-9-anthryl)-4'-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBAPA), N,N''-(2-tert-butylanthracene-9,10-diyl)-4,1-phenylene)bis[N,N',N'-triphenyl-1,4-phenylenediamine](abbreviation: DPABPA), N,9-diphenyl-N-[4-(9,10-diphenyl-2-anthryl)phenyl]-9H-carbazol-3-amine (abbreviation: 2PCAPPA), N-[4-(9,10-diphenyl-2-anthryl)phenyl]-N,N',N'-triphenyl-1,4-phenylenediamine (abbreviation: 2DPAPPA), N,N,N',N',N'',N'',N''',N'''-octaphenyldibenzo[g,p]chrysene-2,7,10,15-tetraamine (abbreviation: DBC1), coumarin 30, N-(9,10-diphenyl-2-anthryl)-N,9-diphenyl-9H-carbazol-3-amine (abbreviation: 2PCAPA), N-[9,10-bis(1,1'-biphenyl-2-yl)-2-anthryl]-N,9-diphenyl-9H-carbazol-3-amine (abbreviation: 2PCABPhA), N-(9,10-diphenyl-2-anthryl)-N,N',N'-triphenyl-1,4-phenylenediamine (abbreviation: 2DPAPA), N-[9,10-bis(1,1'-biphenyl-2-yl)-2-anthryl]-N,N',N'-triphenyl-1,4-phenylenediamine (abbreviation: 2DPA BPhA), 9,10-bis(1,1'-biphenyl-2-yl)-N-[4-(9H-carbazol-9-yl)phenyl]-N-phenylanthracen-2-amine (abbreviation: 2Y GABPhA), N,N,9-triphenylanthracen-9-amine (abbreviation: DPhA PhA) coumarin 545T, N,N'-diphenylquinacridone (abbreviation: DPQd), rubrene, 5,12-bis(1,1'-biphenyl-4-yl)-6,11-diphenyltetracene (abbreviation: BPT), 2-(2-{2-[4-(dimethylamino)phenyl]ethenyl}-6-methyl-4H-pyran-4-ylidene)propanedinitrile (abbreviation: DCM 1), 2-{2-methyl-6-[2-(2,3,6,7-tetrahydro-1H,5H-benzo[ij]quinolizin-9-yl)ethenyl]-4H-pyran-4-ylidene}propanedinitrile (abbreviation: DCM 2), N,N,N',N'-tetrakis(4-methylphenyl)tetracene-5,11-diamine (abbreviation: p-mPhTD), 7,14-diphenyl-N,N,N',N'-tetrakis(4-methylphenyl)acenaphtho[1,2-a]fluoranthene-3,10-diamine (abbreviation: p-mPhAFD), 2-{2-isopropyl-6-[2-(1,1,7,7-tetramethyl-2,3,6,7-tetrahydro-1H,5H-benzo[ij]quinolizin-9-yl)ethenyl]-4H-pyran-4-ylidene}propanedinitrile (abbreviation: DCJTI), 2-{2-tert-butyl-6-[2-(1,1,7,7-tetramethyl-2,3,6,7-tetrahydro-1H,5H-benzo[ij]quinolizin-9-yl)ethenyl]-4H-pyran-4-ylidene}propanedinitrile (abbreviation: DCJTB), 2-(2,6-bis{2-[4-(dimethylamino)phenyl]ethenyl}-4H-pyran-4-ylidene)propanedinitrile (abbreviation: BisDCM), 2-{2,6-bis[2-(8-methoxy-1,1,7,7-tetramethyl-2,3,6,7-tetrahydro-1H,5H-benzo[ij]quinolizin-9-yl)ethenyl]-4H-pyran-4-ylidene}propanedinitrile (abbreviation: BisDCJTM), and N,N'-diphenyl-N,N'-(1,6-pyrene-diyl)bis[(6-phenylbenzo[b]naphtho[1,2-d]furan)-8-amine](abbreviation: 1,6BnfAPrn-03) can be given. In particular, a condensed aromatic diamine compound typified by a pyrenediamine compound such as 1,6FLPAPrn, 1,6mMemFLPAPrn, and 1,6BnfAPrn-03 is preferable because of its high hole-trapping property, high emission efficiency, and high reliability.

[0118] Examples of a material that can be used as a phosphorescent substance in the light-emitting layer **113** are as follows.

[0119] An organometallic iridium complex having a 4H-triazole skeleton, such as tris{2-[5-(2-methylphenyl)-4-(2,6-dimethylphenyl)-4H-1,2,4-triazol-3-yl-κN2]phenyl-KC}iridium(III) (abbreviation: [Ir(mpptz-dmp).sub.3]), tris(5-methyl-3,4-diphenyl-4H-1,2,4-triazolato)iridium(III) (abbreviation: [Ir(Mptz).sub.3]), or tris[4-(3-biphenyl)-5-isopropyl-3-phenyl-4H-1,2,4-triazolato]iridium(III) (abbreviation: [Ir(iPrptz-3b).sub.3]), an organometallic iridium complex having a 1H-triazole skeleton, such as tris[3-methyl-1-(2-methylphenyl)-5-phenyl-1H-1,2,4-triazolato]iridium(III) (abbreviation: [Ir(Mptzl-mp).sub.3]) or tris(1-methyl-5-phenyl-3-propyl-1H-1,2,4-triazolato)iridium(III) (abbreviation: [Ir(Prptzl-Me).sub.3]), an organometallic iridium complex having an imidazole skeleton, such as fac-tris[1-(2,6-diisopropylphenyl)-2-phenyl-1H-imidazole]iridium(III) (abbreviation: [Ir(iPrpmi).sub.3]) or tris[3-(2,6-dimethylphenyl)-7-methylimidazo[1,2-f]phenanthridinato]iridium(III) (abbreviation: [Ir(dmpimpt-Me).sub.3]), and an organometallic iridium complex in which a phenylpyridine derivative having an electron-withdrawing group is a ligand, such as bis[2-(4',6'-difluorophenyl)pyridinato-N,C2']iridium(III) tetrakis(1-pyrazolyl)borate (abbreviation: FIr6), bis[2-(4',6'-difluorophenyl)pyridinato-N,C2']iridium(III) picolinate (abbreviation: Firpic), bis{2-[3',5'-

bis(trifluoromethyl)phenyl]pyridinato-N,C2'}iridium(III) picolinate (abbreviation: [Ir(CF3ppy).sub.2(pic)]), or bis[2-(4',6'-difluorophenyl)pyridinato-N,C2']iridium(III) acetylacetonate (abbreviation: FIracac) can be given. These are compounds exhibiting blue phosphorescence, and are compounds having an emission peak at 440 nm to 520 nm.

[0120] Furthermore, an organometallic iridium complex having a pyrimidine skeleton, such as tris(4-methyl-6-phenylpyrimidinato)iridium(III) (abbreviation: [Ir(mppm).sub.3]), tris(4-*t*-butyl-6-phenylpyrimidinato)iridium(III) (abbreviation: [Ir(tBuppm).sub.3]), (acetylacetonato)bis(6-methyl-4-phenylpyrimidinato)iridium(III) (abbreviation: [Ir(mppm).sub.2(acac)]), (acetylacetonato)bis(6-*tert*-butyl-4-phenylpyrimidinato)iridium(III) (abbreviation: [Ir(tBuppm).sub.2(acac)]), (acetylacetonato)bis[6-(2-norbornyl)-4-phenylpyrimidinato]iridium(III) (abbreviation: [Ir(nbppm).sub.2(acac)]), (acetylacetonato)bis[5-methyl-6-(2-methylphenyl)-4-phenylpyrimidinato]iridium(III) (abbreviation: [Ir(mpmppm).sub.2(acac)]), or (acetylacetonato)bis(4,6-diphenylpyrimidinato)iridium(III) (abbreviation: [Ir(dppm).sub.2(acac)]), an organometallic iridium complex having a pyrazine skeleton, such as (acetylacetonato)bis(3,5-dimethyl-2-phenylpyrazinato)iridium(III) (abbreviation: [Ir(mppr-Me).sub.2(acac)]) or (acetylacetonato)bis(5-isopropyl-3-methyl-2-phenylpyrazinato)iridium(III) (abbreviation: [Ir(mppr-iPr).sub.2(acac)]), an organometallic iridium complex having a pyridine skeleton, such as tris(2-phenylpyridinato-N,C.sup.2')iridium(III) (abbreviation: [Ir(ppy).sub.3]), bis(2-phenylpyridinato-N,C.sup.2')iridium(III) acetylacetonate (abbreviation: [Ir(ppy).sub.2(acac)]), bis(benzo[h]quinolinato)iridium(III) acetylacetonate (abbreviation: [Ir(bzq).sub.2(acac)]), tris(benzo[h]quinolinato)iridium(III) (abbreviation: [Ir(bzq).sub.3]), tris(2-phenylquinolinato-N,C.sup.2')iridium(III) (abbreviation: [Ir(pq).sub.3]), or bis(2-phenylquinolinato-N,C.sup.2')iridium(III) acetylacetonate (abbreviation: [Ir(pq).sub.2(acac)]), and a rare earth metal complex such as tris(acetylacetonato)(monophenanthroline)terbium(III) (abbreviation: [Tb(acac).sub.3(Phen)]) can be given. These are mainly compounds exhibiting green phosphorescence, and have an emission peak at 500 nm to 600 nm. Note that an organometallic iridium complex having a pyrimidine skeleton is particularly preferable because of its distinctively high reliability and emission efficiency.

[0121] Furthermore, an organometallic iridium complex having a pyrimidine skeleton, such as (diisobutyrylmethanato)bis[4,6-bis(3-methylphenyl)pyrimidinato]iridium(III) (abbreviation: [Ir(5mdppm).sub.2(dibm)]), bis[4,6-bis(3-methylphenyl)pyrimidinato](dipivaloylmethanato)iridium(III) (abbreviation: [Ir(5mdppm).sub.2(dpm)]), or bis[4,6-di(naphthalen-1-yl)pyrimidinato](dipivaloylmethanato)iridium(III) (abbreviation: [Ir(dlnpm).sub.2(dpm)]), an organometallic iridium complex having a pyrazine skeleton, such as (acetylacetonato)bis(2,3,5-triphenylpyrazinato)iridium(III) (abbreviation: [Ir(tppr).sub.2(acac)]), bis(2,3,5-triphenylpyrazinato)(dipivaloylmethanato)iridium(III) (abbreviation: [Ir(tppr).sub.2(dpm)]), or (acetylacetonato)bis[2,3-bis(4-fluorophenyl)quinoxalinato]iridium(III) (abbreviation: [Ir(Fdpq).sub.2(acac)]), an organometallic iridium complex having a pyridine skeleton, such as tris(1-phenylisoquinolinato-N,C2')iridium(III) (abbreviation: [Ir(piq).sub.3]) or bis(1-phenylisoquinolinato-N,C2')iridium(III) acetylacetonate (abbreviation: [Ir(piq).sub.2(acac)]), a platinum complex such as 2,3,7,8,12,13,17,18-octaethyl-21H,23H-porphyrin platinum(II) (abbreviation: PtOEP), and a rare earth metal complex such as tris(1,3-diphenyl-1,3-propanedionato)(monophenanthroline)europium(III) (abbreviation: [Eu(DBM).sub.3(Phen)]) or tris[1-(2-thenoyl)-3,3,3-trifluoroacetato](monophenanthroline)europium(III) (abbreviation: [Eu(TTA).sub.3(Phen)]) can be given. These are compounds exhibiting red phosphorescence, and have an emission peak at 600 nm to 700 nm. Furthermore, the organometallic iridium complexes including a pyrazine skeleton can emit red light with favorable chromaticity.

[0122] Besides the above-described phosphorescent compounds, other known phosphorescent materials may be selected and used.

[0123] As the TADF material, a fullerene, a derivative thereof, an acridine, a derivative thereof, an

eosin derivative, or the like can be used. Other examples include a metal-containing porphyrin containing magnesium (Mg), zinc (Zn), cadmium (Cd), tin (Sn), platinum (Pt), indium (In), palladium (Pd), or the like. Examples of the metal-containing porphyrin include a protoporphyrin-tin fluoride complex (SnF.sub.2(Proto IX)), a mesoporphyrin-tin fluoride complex (SnF.sub.2(Meso IX)), a hematoporphyrin-tin fluoride complex (SnF.sub.2(Hemato IX)), a coproporphyrin tetramethyl ester-tin fluoride complex (SnF.sub.2(Copro III-4Me)), an octaethylporphyrin-tin fluoride complex (SnF.sub.2(OEP)), an etioporphyrin-tin fluoride complex (SnF.sub.2(Etio I)), and an octaethylporphyrin-platinum chloride complex (PtCl.sub.2(OEP)), which are expressed by the following structural formulae.

##STR00001## ##STR00002## ##STR00003##

[0124] Alternatively, a heterocyclic compound having both of a 7π -electron rich heteroaromatic ring and a π -electron deficient heteroaromatic ring that is expressed by the following structural formulae, such as 2-(biphenyl-4-yl)-4,6-bis(12-phenylindolo[2,3-a]carbazol-11-yl)-1,3,5-triazine (abbreviation: PIC-TRZ), 9-(4,6-diphenyl-1,3,5-triazin-2-yl)-9'-phenyl-9H,9'H-3,3'-bicarbazole (abbreviation: PCCzTzn), 9-[4-(4,6-diphenyl-1,3,5-triazin-2-yl)phenyl]-9'-phenyl-9H,9'H-3,3'-bicarbazole (abbreviation: PCCzPTzn), 2-[4-(10H-phenoxazine-10-yl)phenyl]-4,6-diphenyl-1,3,5-triazine (abbreviation: PXZ-TRZ), 3-[4-(5-phenyl-5,10-dihydrophenazin-10-yl)phenyl]-4,5-diphenyl-1,2,4-triazole (abbreviation: PPZ-3TPT), 3-(9,9-dimethyl -9H-acridin-10-yl)-9H-xanthen-9-one (abbreviation: ACRXTN), bis[4-(9,9-dimethyl-9,10-dihydroacridine)phenyl]sulfone (abbreviation: DMAC-DPS), or 10-phenyl-10H,10'H-spiro[acridin-9,9'-anthracen]-10'-one (abbreviation: ACRSA) can be used. The heterocyclic compound is preferable because of having both a high electron-transport property and a high hole-transport property owing to a π -electron rich heteroaromatic ring and a π -electron deficient heteroaromatic ring. Note that a substance in which the π -electron rich heteroaromatic ring and the π -electron deficient heteroaromatic ring are directly bonded to each other is particularly preferable because the donor property of the π -electron rich heteroaromatic ring and the acceptor property of the π -electron deficient heteroaromatic ring are both increased and the energy difference between the S1 level and the T1 level becomes small, so that thermally activated delayed fluorescence can be obtained with high efficiency. Note that an aromatic ring to which an electron-withdrawing group such as a cyano group is bonded may be used instead of the π -electron deficient heteroaromatic ring.

##STR00004## ##STR00005##

[0125] As the host material in the light-emitting layer, a variety of carrier-transport materials such as a material having an electron-transport property and a material having a hole-transport property can be used.

[0126] As the material having a hole-transport property, the substance given as the material having a hole-transport property which is contained in the hole-transport layer **112** can be favorably used.

[0127] As the material having an electron-transport property, for example, a metal complex such as bis(10-hydroxybenzo[h]quinolinato)beryllium(II) (abbreviation: BeBq.sub.2), bis(2-methyl-8-quinolinolato)(4-phenylphenolato)aluminum(II) (abbreviation: BALq), bis(8-quinolinolato)zinc(II) (abbreviation: Znq), bis[2-(2-benzoxazolyl)phenolato]zinc(II) (abbreviation: ZnPBO), or bis[2-(2-benzothiazolyl)phenolato]zinc(II) (abbreviation: ZnBTZ), a heterocyclic compound having a polyazole skeleton, such as 2-(4-biphenyl)-5-(4-tert-butylphenyl)-1,3,4-oxadiazole (abbreviation: PBD), 3-(4-biphenyl)-4-phenyl-5-(4-tert-butylphenyl)-1,2,4-triazole (abbreviation: TA Z), 1,3-bis[5-(p-tert-butylphenyl)-1,3,4-oxadiazol-2-yl]benzene (abbreviation: OX D-7), 9-[4-(5-phenyl-1,3,4-oxadiazol-2-yl)phenyl]-9H-carbazole (abbreviation: C011), 2,2', 2''-(1,3,5-benzenetriyl)tris(1-phenyl-1H-benzimidazole) (abbreviation: TPB!), or 2-[3-(dibenzothiophen-4-yl)phenyl]-1-phenyl-1H-benzimidazole (abbreviation: mDBTBIIm-II), a heterocyclic compound having a diazine skeleton, such as 2-[3-(dibenzothiophen-4-yl)phenyl]dibenzo[f,h]quinoxaline (abbreviation: 2mDBTPDBq-II), 2-[3'-(dibenzothiophen-4-yl)biphenyl-3-yl]dibenzo[f,h]quinoxaline (abbreviation: 2mDBTBPDBq-II), 2-[3'-(9H-carbazol-9-yl)biphenyl-3-

yl]dibenzo[f,h]quinoxaline (abbreviation: 2mCzBPDBq), 4,6-bis[3-(phenanthren-9-yl)phenyl]pyrimidine (abbreviation: 4,6mPnP2Pm), or 4,6-bis[3-(4-dibenzothienyl)phenyl]pyrimidine (abbreviation: 4,6mDBTP2Pm-II), and a heterocyclic compound having a pyridine skeleton, such as 3,5-bis[3-(9H-carbazol-9-yl)phenyl]pyridine (abbreviation: 35DCzPPy) or 1,3,5-tri[3-(3-pyridyl)-phenyl]benzene (abbreviation: TmPyPB) can be given. Among the above, the heterocyclic compound having a diazine skeleton and the heterocyclic compound having a pyridine skeleton have favorable reliability and thus are preferable. In particular, the heterocyclic compound having a diazine (pyrimidine or pyrazine) skeleton has a high electron-transport property and contributes to a reduction in driving voltage.

[0128] In the case where a fluorescent substance is used as the light-emitting material, a material having an anthracene skeleton is suitable for the host material. The use of a substance having an anthracene skeleton as a host material for a fluorescent substance makes it possible to achieve a light-emitting layer with favorable emission efficiency and durability. Most of materials having an anthracene skeleton have a deep HOMO level; therefore, such a material can be preferably used in one embodiment of the present invention. As the substance having an anthracene skeleton that is used as the host material, a substance having a diphenylanthracene skeleton, in particular, a substance having a 9,10-di phenylanthracene skeleton, is preferable because of its chemical stability. The host material preferably has a carbazole skeleton because the hole-injection and hole-transport properties are improved; further preferably, the host material has a benzocarbazole skeleton in which a benzene ring is further condensed to carbazole because the HOMO level thereof is shallower than that of carbazole by approximately 0.1 eV and thus holes enter the host material easily. In particular, the host material having a dibenzocarbazole skeleton is preferable because its HOMO level is shallower than that of carbazole by approximately 0.1 eV so that holes enter the host material easily, the hole-transport property is improved, and the heat resistance is increased. Accordingly, a substance that has both a 9,10-di phenylanthracene skeleton and a carbazole skeleton (or a benzocarbazole skeleton or a dibenzocarbazole skeleton) is further preferable as the host material. Note that in terms of the hole-injection and hole-transport properties described above, instead of a carbazole skeleton, a benzofluorene skeleton or a dibenzo fluorene skeleton may be used. Examples of such a substance include 9-phenyl-3-[4-(10-phenyl-9-anthryl)phenyl]-9H-carbazole (abbreviation: PCzPA), 3-[4-(1-naphthyl)-phenyl]-9-phenyl-9H-carbazole (abbreviation: PCPN), 9-[4-(10-phenyl-9-anthracenyl)phenyl]-9H-carbazole (abbreviation: CzPA), 7-[4-(10-phenyl-9-anthryl)phenyl]-7H-dibenzo[c,g]carbazole (abbreviation: cgDBCzPA), 6-[3-(9,10-diphenyl-2-anthryl)phenyl]-benzo[b]naphtho[1,2-d]furan (abbreviation: 2mBnfPPA), and 9-phenyl-10-{4-(9-phenyl-9H-fluoren-9-yl)biphenyl-4'-yl}anthracene (abbreviation: FLPPA). In particular, CzPA, cgDBCzPA, 2mBnfPPA, and PCzPA are preferably selected because they exhibit favorable characteristics.

[0129] Note that a host material may be a material of a mixture of a plurality of kinds of substances; in the case of using a mixed host material, it is preferable to mix a material having an electron-transport property with a material having a hole-transport property. When the material having an electron-transport property is mixed with the material having a hole-transport property, the transport property of the light-emitting layer **113** can be easily adjusted and a recombination region can be easily controlled. The ratio of the content of the material having a hole-transport property to the content of the material having an electron-transport property may be the material having a hole-transport property: the material having an electron-transport property=1:9 to 9:1.

[0130] An exciplex may be formed by these mixed materials. A combination is preferably selected so as to form an exciplex that exhibits light emission overlapping with the wavelength of a lowest-energy-side absorption band of a light-emitting material, because energy can be transferred smoothly and light emission can be efficiently obtained. The use of the structure is preferable because the driving voltage is also be reduced.

[0131] It is particularly suitable for a blue light emitting device in one embodiment of the present

invention.

[0132] The electron-transport layer **114** is a layer containing a substance having an electron-transport property. As the substance having an electron-transport property, it is possible to use any of the above-listed substances having electron-transport properties that can be used as the host material.

[0133] As the electron-injection layer **115**, a layer containing an alkali metal, an alkaline earth metal, or a compound thereof, such as lithium fluoride (LiF), cesium fluoride (CsF), or calcium fluoride (CaF₂), may be provided between the electron-transport layer **114** and the second electrode **102**. For example, an electride or a layer that is formed using a substance having an electron-transport property and that includes an alkali metal, an alkaline earth metal, or a compound thereof can be used as the electron-injection layer **115**. Examples of the electride include a substance in which electrons are added at high concentration to a mixed oxide of calcium and aluminum.

[0134] Note that as the electron-injection layer **115**, it is possible to use a layer that contains a substance having an electron-transport property (preferably an organic compound having a bipyridine skeleton) and contains a fluoride of the alkali metal or the alkaline earth metal at a concentration higher than that at which the electron-injection layer **115** becomes in a microcrystalline state (50 wt % or higher). Since the layer has a low refractive index, a light-emitting device including the layer can have high external quantum efficiency.

[0135] Instead of the electron-injection layer **115**, a charge-generation layer **116** may be provided (FIG. 4(B)). The charge-generation layer **116** refers to a layer capable of injecting holes into a layer in contact therewith on the cathode side and injecting electrons into a layer in contact therewith on the anode side when supplied with a potential. The charge-generation layer **116** includes at least a P-type layer **117**. The P-type layer **117** is preferably formed using the composite materials given above as the material that can form the hole-injection layer **111**. The P-type layer **117** may be formed by stacking a film containing the above acceptor material as a material included in the composite material and a film containing the above hole-transport material. When a potential is applied to the P-type layer **117**, electrons are injected into the electron-transport layer **114** and holes are injected into the second electrode **102** that is a cathode; thus, the light-emitting device operates. Since the organic compound of one embodiment of the present invention has a low refractive index, using the organic compound for the P-type layer **117** enables the light-emitting device to have high external quantum efficiency.

[0136] Note that one or both of an electron-relay layer **118** and an electron-injection buffer layer **119** are preferably provided in the charge-generation layer **116** in addition to the P-type layer **117**.

[0137] The electron-relay layer **118** contains at least a substance having an electron-transport property and has a function of preventing an interaction between the electron-injection buffer layer **119** and the P-type layer **117** to transfer electrons smoothly. The LUMO level of the substance having an electron-transport property contained in the electron-relay layer **118** is preferably between the LUMO level of an acceptor substance in the P-type layer **117** and the LUMO level of a substance contained in a layer of the electron-transport layer **114** in contact with the charge-generation layer **116**. A specific energy level of the LUMO level of the substance having an electron-transport property used for the electron-relay layer **118** may be higher than or equal to -5.0 eV, preferably higher than or equal to -5.0 eV and lower than or equal to -3.0 eV. Note that as the substance having an electron-transport property used for the electron-relay layer **118**, a phthalocyanine-based material or a metal complex having a metal-oxygen bond and an aromatic ligand is preferably used.

[0138] For the electron-injection buffer layer **119**, a substance having a high electron-injection property, such as an alkali metal, an alkaline earth metal, a rare earth metal, or a compound thereof (an alkali metal compound (including an oxide such as lithium oxide, a halide, and a carbonate such as lithium carbonate or cesium carbonate), an alkaline earth metal compound (including an

oxide, a halide, and a carbonate), or a rare earth metal compound (including an oxide, a halide, and a carbonate)), can be used.

[0139] In the case where the electron-injection buffer layer **119** is formed so as to contain the substance having an electron-transport property and a donor substance, an organic compound such as tetrathianaphthacene (abbreviation: TTN), nickelocene, or decamethylnickelocene can be used as the donor substance, as well as an alkali metal, an alkaline earth metal, a rare earth metal, a compound thereof (an alkali metal compound (including an oxide such as lithium oxide, a halide, and a carbonate such as lithium carbonate or cesium carbonate), an alkaline earth metal compound (including an oxide, a halide, and a carbonate), or a rare earth metal compound (including an oxide, a halide, and a carbonate)). Note that as the substance having an electron-transport property, a material similar to the above-described material forming the electron-transport layer **114** can be used for the formation.

[0140] As a substance forming the second electrode **102**, a metal, an alloy, an electrically conductive compound, or a mixture thereof having a low work function (specifically, 3.8 eV or less) or the like can be used. As specific examples of such a cathode material, elements belonging to Group 1 or Group 2 of the periodic table, such as alkali metals, e.g., lithium (Li) and cesium (Cs), magnesium (Mg), calcium (Ca), and strontium (Sr), alloys containing these (MgAg and AlLi), rare earth metals such as europium (Eu) and ytterbium (Yb), alloys containing these rare earth metals, and the like can be given. However, when the electron-injection layer is provided between the second electrode **102** and the electron-transport layer, as the second electrode **102**, a variety of conductive materials such as Al, Ag, ITO, or indium oxide-tin oxide containing silicon or silicon oxide can be used regardless of their work functions. Films of these conductive materials can be formed by a dry process such as a vacuum evaporation method or a sputtering method, an inkjet method, a spin coating method, or the like. Alternatively, the films may be formed by a wet process using a sol-gel method or a wet process using a paste of a metal material.

[0141] Various methods can be used as a method for forming the EL layer **103** regardless of whether it is a dry process or a wet process. For example, a vacuum evaporation method, a gravure printing method, an offset printing method, a screen printing method, an ink-jet method, a spin coating method, or the like may be used.

[0142] Different deposition methods may be used to form the electrodes or the layers described above.

[0143] The structure of the layers provided between the first electrode **101** and the second electrode **102** is not limited to the above-described structure. However, a structure is preferable in which a light-emitting region where holes and electrons recombine is provided at a position away from the first electrode **101** and the second electrode **102** so as to prevent quenching caused by the proximity of the light-emitting region and a metal used for electrodes and carrier-injection layers.

[0144] Furthermore, in order to inhibit energy transfer from an exciton generated in the light-emitting layer, it is preferable to form the hole-transport layer and the electron-transport layer that are in contact with the light-emitting layer **113**, particularly a carrier-transport layer closer to the recombination region in the light-emitting layer **113**, using the light-emitting material of the light-emitting layer or a substance having a wider band gap than the light-emitting material included in the light-emitting layer.

Embodiment 2

[0145] In this embodiment, a light-emitting apparatus using the light-emitting device described in Embodiment 1 will be described.

[0146] In this embodiment, a light-emitting apparatus fabricated using the light-emitting device described in Embodiment 1 will be described with reference to FIG. 5. Note that FIG. 5(A) is a top view illustrating the light-emitting apparatus and FIG. 5(B) is a cross-sectional view taken along A-B and C-D in FIG. 5(A). This light-emitting apparatus includes a driver circuit portion (source line driver circuit) **601**, a pixel portion **602**, and a driver circuit portion (gate line driver circuit) **603**,

which are for controlling light emission of a light-emitting device and are illustrated with dotted lines. Furthermore, **604** denotes a sealing substrate, **605** denotes a sealant, and the inside surrounded by the sealant **605** is a space **607**.

[0147] Note that a lead wiring **608** is a wiring for transmitting signals to be input to the source line driver circuit portion **601** and the gate line driver circuit **603** and receiving a video signal, a clock signal, a start signal, a reset signal, and the like from an FPC (flexible printed circuit) **609** serving as an external input terminal. Although only the FPC is illustrated here, a printed wiring board (PWB) may be attached to this FPC. The light-emitting apparatus in this specification includes not only the light-emitting apparatus itself but also the apparatus provided with the FPC or the PWB.

[0148] Next, a cross-sectional structure is described with reference to FIG. 5(B). The driver circuit portion and the pixel portion are formed over an element substrate **610**. Here, the source line driver circuit portion **601**, which is the driver circuit portion, and one pixel of the pixel portion **602** are illustrated.

[0149] The element substrate **610** may be fabricated using a substrate containing glass, quartz, an organic resin, a metal, an alloy, a semiconductor, or the like, or a plastic substrate formed of FRP (Fiber Reinforced Plastic), PVF (polyvinyl fluoride), polyester, an acrylic resin, or the like.

[0150] The structure of transistors used in pixels and driver circuits is not particularly limited. For example, an inverted staggered transistor or a staggered transistor may be used. Furthermore, top-gate transistors or bottom-gate transistors may be used. A semiconductor material used for the transistors is not particularly limited, and for example, silicon, germanium, silicon carbide, gallium nitride, or the like can be used. Alternatively, an oxide semiconductor containing at least one of indium, gallium, and zinc, such as In—Ga—Zn-based metal oxide, may be used.

[0151] There is no particular limitation on the crystallinity of a semiconductor material used for the transistor, and an amorphous semiconductor or a semiconductor having crystallinity (a microcrystalline semiconductor, a polycrystalline semiconductor, a single crystal semiconductor, or a semiconductor partly including crystal regions) can be used. A semiconductor having crystallinity is preferably used, in which case deterioration of the transistor characteristics can be suppressed.

[0152] Here, an oxide semiconductor is preferably used for semiconductor devices such as the transistors provided in the pixels and driver circuits and transistors used for touch sensors described later, and the like. In particular, an oxide semiconductor having a wider band gap than silicon is preferably used. The use of an oxide semiconductor material having a wider band gap than silicon can reduce the off-state current of the transistors.

[0153] The oxide semiconductor preferably contains at least indium (In) or zinc (Zn). Further preferably, the oxide semiconductor contains an oxide represented by an In-M—Zn-based oxide (M represents a metal such as Al, Ti, Ga, Ge, Y, Zr, Sn, La, Ce, or Hf).

[0154] As a semiconductor layer, it is particularly preferable to use an oxide semiconductor film including a plurality of crystal parts whose c-axes are aligned perpendicular to a surface on which the semiconductor layer is formed or the top surface of the semiconductor layer and in which the adjacent crystal parts have no grain boundary.

[0155] The use of such a material as the semiconductor layer makes it possible to achieve a highly reliable transistor in which a change in the electrical characteristics is reduced.

[0156] Charge accumulated in a capacitor through a transistor including the above-described semiconductor layer can be retained for a long time because of the low off-state current of the transistor. The use of such a transistor in pixels allows a driver circuit to stop while the gray level of an image displayed on each display region is maintained. As a result, an electronic device with significantly reduced power consumption can be achieved.

[0157] For stable characteristics of the transistor or the like, a base film is preferably provided. The base film can be formed to be a single-layer or a stacked-layer using an inorganic insulating film such as a silicon oxide film, a silicon nitride film, a silicon oxynitride film, or a silicon nitride oxide film. The base film can be formed by a sputtering method, a CVD (Chemical Vapor

Deposition) method (e.g., a plasma CVD method, a thermal CVD method, or an MOCVD (Metal Organic CVD) method), an ALD (Atomic Layer Deposition) method, a coating method, a printing method, or the like. Note that the base film is not necessarily provided when not needed.

[0158] Note that an FE T **623** is illustrated as a transistor formed in the driver circuit portion **601**. The driver circuit can be formed using various circuits such as a CMOS circuit, a PMOS circuit, and an NMOS circuit. Although a driver-integrated type in which the driver circuit is formed over the substrate is described in this embodiment, the driver circuit is not necessarily formed over the substrate and can be formed outside.

[0159] The pixel portion **602** is formed with a plurality of pixels including a switching FET **611**, a current control FET **612**, and a first electrode **613** electrically connected to a drain of the current control FET **612**; however, without being limited thereto, a pixel portion in which three or more FETs and a capacitor are combined may be employed.

[0160] Note that an insulator **614** is formed to cover an end portion of the first electrode **613**. The insulator **614** can be formed using a positive photosensitive acrylic resin film here.

[0161] In order to improve the coverage with an EL layer or the like to be formed later, the insulator **614** is formed so as to have a curved surface with curvature at its upper end portion or lower end portion. For example, in the case where a positive photosensitive acrylic resin is used as a material for the insulator **614**, only the upper end portion of the insulator **614** preferably has a curved surface with a curvature radius (0.2 μm to 3 μm). As the insulator **614**, either a negative photosensitive resin or a positive photosensitive resin can be used.

[0162] An EL layer **616** and a second electrode **617** are formed over the first electrode **613**. Here, as a material used for the first electrode **613** functioning as an anode, a material with a high work function is desirably used. For example, a single-layer film of an ITO film, an indium tin oxide film containing silicon, an indium oxide film containing zinc oxide at 2 wt % to 20 wt %, a titanium nitride film, a chromium film, a tungsten film, a Zn film, a Pt film, or the like, a stacked layer of titanium nitride film and a film containing aluminum as its main component, a three-layer structure of a titanium nitride film, a film containing aluminum as its main component, and a titanium nitride film, or the like can be used. Note that the stacked-layer structure achieves low wiring resistance, a favorable ohmic contact, and a function as an anode.

[0163] The EL layer **616** is formed by any of a variety of methods such as an evaporation method using an evaporation mask, an inkjet method, and a spin coating method. The EL layer **616** has the structure described in Embodiment 1. Alternatively, a material included in the EL layer **616** may be a low molecular compound or a high molecular compound (including an oligomer or a dendrimer).

[0164] As a material used for the second electrode **617**, which is formed over the EL layer **616** and functions as a cathode, a material with a low work function (e.g., Al, Mg, Li, Ca, or an alloy or a compound thereof (e.g., MgAg, MgIn, or AlLi)) is preferably used. Note that in the case where light generated in the EL layer **616** passes through the second electrode **617**, it is preferable to use, for the second electrode **617**, a stacked layer of a thin metal film and a transparent conductive film (e.g., ITO, indium oxide containing zinc oxide at 2 wt % to 20 wt %, indium tin oxide containing silicon, or zinc oxide (ZnO)).

[0165] Note that a light-emitting device is formed with the first electrode **613**, the EL layer **616**, and the second electrode **617**. The light-emitting device is the light-emitting device described in Embodiment 1. A plurality of light-emitting devices are formed in the pixel portion, and the light-emitting apparatus of this embodiment may include both the light-emitting device described in Embodiment 1 and a light-emitting device having a different structure.

[0166] The sealing substrate **604** and the element substrate **610** are attached to each other using the sealant **605**, so that a structure in which the light-emitting device **618** is provided in the space **607** surrounded by the element substrate **610**, the sealing substrate **604**, and the sealant **605** is employed. The space **607** is filled with a filler; it is filled with an inert gas (e.g., nitrogen or argon) in some cases, and filled with the sealant in some cases. The structure of the sealing substrate in

which a recessed portion is formed and a desiccant is provided is preferable because deterioration due to the influence of moisture can be inhibited.

[0167] Note that an epoxy-based resin or glass frit is preferably used for the sealant **605**.

Furthermore, these materials are preferably materials that transmit as little moisture or oxygen as possible. As a material used for the sealing substrate **604**, in addition to a glass substrate and a quartz substrate, a plastic substrate formed of FRP (Fiber Reinforced Plastics), PVF (polyvinyl fluoride), polyester, an acrylic resin, or the like can be used.

[0168] Although not illustrated in FIG. 5, a protective film may be provided over the second electrode. The protective film may be formed using an organic resin film or an inorganic insulating film. The protective film may be formed so as to cover an exposed portion of the sealant **605**. The protective film may be provided so as to cover surfaces and side surfaces of the pair of substrates and exposed side surfaces of a sealing layer, an insulating layer, and the like.

[0169] For the protective film, a material that is less likely to transmit an impurity such as water. Thus, diffusion of an impurity such as water from the outside into the inside can be effectively inhibited.

[0170] As a material included in the protective film, an oxide, a nitride, a fluoride, a sulfide, a ternary compound, a metal, a polymer, or the like can be used; for example, it is possible to use a material containing aluminum oxide, hafnium oxide, hafnium silicate, lanthanum oxide, silicon oxide, strontium titanate, tantalum oxide, titanium oxide, zinc oxide, niobium oxide, zirconium oxide, tin oxide, yttrium oxide, cerium oxide, scandium oxide, erbium oxide, vanadium oxide, indium oxide; a material containing aluminum nitride, hafnium nitride, silicon nitride, tantalum nitride, titanium nitride, niobium nitride, molybdenum nitride, zirconium nitride, gallium nitride; a material containing a nitride containing titanium and aluminum, an oxide containing titanium and aluminum, an oxide containing aluminum and zinc, a sulfide containing manganese and zinc, a sulfide containing cerium and strontium, an oxide containing erbium and aluminum, an oxide containing yttrium and zirconium, or the like.

[0171] The protective film is preferably formed using a deposition method that enables favorable step coverage. One such method is an atomic layer deposition (ALD) method. A material that can be formed by an ALD method is preferably used for the protective film. With use of an ALD method, a dense protective film with reduced defects such as cracks and pinholes or with a uniform thickness can be formed. Furthermore, damage caused to a process member in forming the protective film can be reduced.

[0172] By an ALD method, a uniform protective film with few defects can be formed even on a surface with a complex uneven shape or upper, side, and lower surfaces of a touch panel.

[0173] As described above, the light-emitting apparatus fabricated using the light-emitting device described in Embodiment 1 can be obtained.

[0174] For the light-emitting apparatus in this embodiment, the light-emitting device described in Embodiment 1 is used and thus a light-emitting apparatus having favorable characteristics can be obtained. Specifically, since the light-emitting device described in Embodiment 1 has high emission efficiency, the light-emitting apparatus can achieve low power consumption.

[0175] This embodiment can be freely combined with any of the other embodiments.

Embodiment 3

[0176] In this embodiment, an example in which the light-emitting device described in Embodiment 1 is used for a lighting device will be described with reference to FIG. 6. FIG. 6(B) is a top view of the lighting device, and FIG. 6(A) is a cross-sectional view taken along e-f in FIG. 6(B).

[0177] In the lighting device in this embodiment, a first electrode **401** is formed over a substrate **400** which is a support and has a light-transmitting property. The first electrode **401** corresponds to the first electrode **101** in Embodiment 1. In the case where light emission is extracted from the first electrode **401** side, the first electrode **401** is formed with a material having a light-transmitting

property.

[0178] A pad **412** for supplying a voltage to a second electrode **404** is formed over the substrate **400**.

[0179] An EL layer **403** is formed over the first electrode **401**. The structure of the EL layer **403** corresponds to, for example, the structure of the EL layer **103** in Embodiment 1. Note that for these structures, the corresponding description can be referred to.

[0180] The second electrode **404** is formed to cover the EL layer **403**. The second electrode **404** corresponds to the second electrode **102** in Embodiment 1. In the case where light-emission is extracted from the first electrode **401** side, the second electrode **404** is formed with a material having high reflectivity. The second electrode **404** is supplied with a voltage when connected to the pad **412**.

[0181] As described above, the lighting device described in this embodiment includes a light-emitting device including the first electrode **401**, the EL layer **403**, and the second electrode **404**. Since the light-emitting device is a light-emitting device with high emission efficiency, the lighting device in this embodiment can be a lighting device with low power consumption.

The substrate **400** over which the light-emitting device having the above structure is formed is fixed to a sealing substrate **407** with sealants **405** and **406** and sealing is performed, whereby the lighting device is completed. It is possible to use only either the sealant **405** or **406**. In addition, the inner sealant **406** (not illustrated in FIG. 6(B)) can be mixed with a desiccant, which enables moisture to be adsorbed, resulting in improved reliability.

[0182] When parts of the pad **412** and the first electrode **401** are provided to extend to the outside of the sealants **405** and **406**, those can serve as external input terminals. An IC chip **420** mounted with a converter or the like may be provided over the external input terminals.

[0183] As described above, the lighting device described in this embodiment uses the light-emitting device described in Embodiment 1 as an EL element, and thus can be a light-emitting apparatus with low power consumption.

Embodiment 4

[0184] In this embodiment, examples of electronic devices each partly including the light-emitting device described in Embodiment 1 are described. The light-emitting device described in Embodiment 1 has high emission efficiency and low power consumption. Thus, the electronic devices described in this embodiment can be electronic devices including light-emitting portions with reduced power consumption.

[0185] Examples of electronic devices to which the light-emitting device is applied include a television devices (also referred to as TV or television receivers), monitors for computers and the like, cameras such as digital cameras and digital video cameras, digital photo frames, mobile phones (also referred to as portable telephones or portable telephone devices), portable game machines, portable information terminals, audio playback devices, and large game machines such as pin-ball machines. Specific examples of these electronic devices are shown below.

[0186] FIG. 7(A) illustrates an example of a television device. In the television device, a display portion **7103** is incorporated in a housing **7101**. Here, a structure in which the housing **7101** is supported by a stand **7105** is shown. Images can be displayed on the display portion **7103**, and the light-emitting devices described in Embodiment 1 are arranged in a matrix in the display portion **7103**.

[0187] The television device can be operated with an operation switch of the housing **7101** or a separate remote controller **7110**. With operation keys **7109** of the remote controller **7110**, channels and volume can be operated and images displayed on the display portion **7103** can be operated. Furthermore, a structure may be employed in which the remote controller **7110** is provided with a display portion **7107** for displaying data output from the remote controller **7110**.

[0188] Note that the television device has a structure of including a receiver, a modem, and the like. With use of the receiver, a general television broadcast can be received, and moreover, when

the television device is connected to a communication network with or without a wire via the modem, one-way (from a sender to a receiver) or two-way (between a sender and a receiver or between receivers) data communication can be performed.

[0189] FIG. 7(B1) is a computer which includes a main body **7201**, a housing **7202**, a display portion **7203**, a keyboard **7204**, an external connection port **7205**, a pointing device **7206**, and the like. Note that this computer is fabricated using the light-emitting devices described in Embodiment 1 arranged in a matrix in the display portion **7203**. The computer in FIG. 7(B2) is provided with a second display portion **7210** instead of the keyboard **7204** and the pointing device **7206**. The second display portion **7210** is of a touch-panel type, and input can be performed by operating display for input displayed on the second display portion **7210** with a finger or a dedicated pen. The second display portion **7210** can also display images other than the display for input. The display portion **7203** may also be a touch panel. Connecting the two screens with a hinge can prevent troubles such as a crack in or damage to the screens caused when the computer is stored or carried.

[0190] FIG. 7(C) illustrates an example of a portable terminal. A mobile phone includes operation buttons **7403**, an external connection port **7404**, a speaker **7405**, a microphone **7406**, and the like in addition to a display portion **7402** incorporated in a housing **7401**. Note that a mobile phone **7400** includes the display portion **7402** which is fabricated by arranging the light-emitting devices described in Embodiment 1 in a matrix.

[0191] The portable terminal illustrated in FIG. 7(C) may have a structure in which information can be input by touching the display portion **7402** with a finger or the like. In this case, operations such as making a call and creating an e-mail can be performed by touching the display portion **7402** with a finger or the like.

[0192] The display portion **7402** has mainly three screen modes. The first one is a display mode mainly for displaying images, and the second one is an input mode mainly for inputting data such as text. The third one is a display+input mode in which two modes of the display mode and the input mode are combined.

[0193] For example, in the case of making a call or creating an e-mail, a text input mode mainly for inputting text is selected for the display portion **7402** so that an operation of inputting text displayed on the screen may be performed. In this case, it is preferable to display a keyboard or number buttons on almost the entire screen of the display portion **7402**.

[0194] When a sensing device including a sensor for sensing inclination, such as a gyroscope sensor or an acceleration sensor, is provided inside the portable terminal, screen display of the display portion **7402** can be automatically changed by determining the orientation of the portable terminal (vertically or horizontally).

[0195] The screen modes are changed by touching the display portion **7402** or operating the operation buttons **7403** of the housing **7401**. Alternatively, the screen modes can be changed depending on the kind of image displayed on the display portion **7402**. For example, when a signal of an image displayed on the display portion is moving image data, the screen mode is changed to the display mode, and when the signal is text data, the screen mode is changed to the input mode.

[0196] Moreover, in the input mode, when input by the touch operation of the display portion **7402** is not performed for a certain period while a signal sensed by an optical sensor in the display portion **7402** is sensed, the screen mode may be controlled so as to be changed from the input mode to the display mode.

[0197] The display portion **7402** can also function as an image sensor. For example, an image of a palm print, a fingerprint, or the like is taken when the display portion **7402** is touched with the palm or the finger, whereby personal authentication can be performed. Furthermore, by using a backlight which emits near-infrared light or a sensing light source which emits near-infrared light in the display portion, an image of a finger vein, a palm vein, or the like can be taken.

[0198] Note that the structures described in this embodiment can be combined with the structures

described in any of Embodiment 1 to Embodiment 3 as appropriate.

[0199] As described above, the application range of the light-emitting apparatus including the light-emitting device described in Embodiment 1 is wide so that this light-emitting apparatus can be applied to electronic devices in a variety of fields. By using the light-emitting device described in Embodiment 1, an electronic device having low power consumption can be obtained.

[0200] FIG. 8(A) is a schematic view illustrating an example of a cleaning robot.

[0201] A cleaning robot **5100** includes a display **5101** placed on its top surface, a plurality of cameras **5102** placed on its side surface, a brush **5103**, and operation buttons **5104**. Although not illustrated, the bottom surface of the cleaning robot **5100** is provided with a tire, an inlet, and the like. Furthermore, the cleaning robot **5100** includes various sensors such as an infrared sensor, an ultrasonic sensor, an acceleration sensor, a piezoelectric sensor, an optical sensor, and a gyroscope sensor. In addition, the cleaning robot **5100** has a wireless communication means.

[0202] The cleaning robot **5100** is self-propelled, detects dust **5120**, and sucks up the dust through the inlet provided on the bottom surface.

[0203] The cleaning robot **5100** can judge whether there is an obstacle such as a wall, furniture, or a step by analyzing images taken by the cameras **5102**. When an object that is likely to be caught in the brush **5103**, such as a wire, is detected by image analysis, the rotation of the brush **5103** can be stopped.

[0204] The display **5101** can display the remaining capacity of a battery, the amount of vacuumed dust, and the like. The display **5101** may display a path on which the cleaning robot **5100** has run. The display **5101** may be a touch panel, and the operation buttons **5104** may be provided on the display **5101**.

[0205] The cleaning robot **5100** can communicate with a portable electronic device **5140** such as a smartphone. The portable electronic device **5140** can display images taken by the cameras **5102**. Accordingly, an owner of the cleaning robot **5100** can monitor the room even from the outside. The display on the display **5101** can be checked by the portable electronic device **5140** such as a smartphone.

[0206] The light-emitting apparatus of one embodiment of the present invention can be used for the display **5101**.

[0207] A robot **2100** illustrated in FIG. 8(B) includes an arithmetic device **2110**, an illuminance sensor **2101**, a microphone **2102**, an upper camera **2103**, a speaker **2104**, a display **2105**, a lower camera **2106**, an obstacle sensor **2107**, and a moving mechanism **2108**.

[0208] The microphone **2102** has a function of detecting a speaking voice of a user, an environmental sound, and the like. The speaker **2104** also has a function of outputting sound. The robot **2100** can communicate with a user using the microphone **2102** and the speaker **2104**.

[0209] The display **2105** has a function of displaying various kinds of information. The robot **2100** can display information desired by a user on the display **2105**. The display **2105** may be provided with a touch panel. Moreover, the display **2105** may be a detachable information terminal, in which case charging and data communication can be performed when the display **2105** is set at the home position of the robot **2100**.

[0210] The upper camera **2103** and the lower camera **2106** each have a function of taking an image of the surroundings of the robot **2100**. The obstacle sensor **2107** can detect the presence of an obstacle in the direction where the robot **2100** advances with the moving mechanism **2108**. The robot **2100** can move safely by recognizing the surroundings with the upper camera **2103**, the lower camera **2106**, and the obstacle sensor **2107**. The light-emitting apparatus of one embodiment of the present invention can be used for the display **2105**.

[0211] FIG. 8(C) shows an example of a goggle-type display. The goggle-type display includes, for example, a housing **5000**, a display portion **5001**, a speaker **5003**, an LED lamp **5004**, an operation keys **5005** (including a power switch or an operation switch), a connection terminal **5006**, a sensor **5007** (a sensor having a function of measuring force, displacement, position, speed, acceleration,

angular velocity, rotational frequency, distance, light, liquid, magnetism, temperature, chemical substance, sound, time, hardness, electric field, current, voltage, power, radiation, flow rate, humidity, gradient, oscillation, odor, or infrared ray), a microphone **5008**, a second display portion **5002**, a support **5012**, an earphone **5013**, and the like.

[0212] The light-emitting apparatus of one embodiment of the present invention can be used for the display portion **5001** and the second display portion **5002**.

[0213] FIG. **9** illustrates an example in which the light-emitting device described in Embodiment 1 is used for a table lamp which serves as a lighting device. The table lamp illustrated in FIG. **9** includes a housing **2001** and a light source **2002**, and the lighting device described in Embodiment 2 may be used for the light source **2002**.

[0214] FIG. **10** illustrates an example in which the light-emitting device described in Embodiment 1 is used for an indoor lighting device **3001**. Since the light-emitting device described in Embodiment 1 has high emission efficiency, a lighting device having low power consumption can be obtained. Furthermore, the light-emitting device described in Embodiment 1 can have a larger area, and thus can be used for a large-area lighting device. Furthermore, the light-emitting device described in Embodiment 1 is thin, and thus can be used for a lighting device having a reduced thickness.

[0215] The light-emitting device described in Embodiment 1 can also be used for a windshield or a dashboard of an automobile. FIG. **11** illustrates one mode in which the light-emitting device described in Embodiment 1 is used for an automobile windshield and an automobile dashboard. A display region **5200** to a display region **5203** are each a display region provided using the light-emitting device described in Embodiment 1.

[0216] The display region **5200** and the display region **5201** are display devices into which the light-emitting devices described in Embodiment 1 are incorporated. When the light-emitting devices described in Embodiment 1 are fabricated using electrodes having light-transmitting properties as a first electrode and a second electrode, what is called see-through display devices, through which the opposite side can be seen, can be obtained. See-through display devices can be provided without hindering the vision even when being provided in the automobile windshield. Note that in the case where a driving transistor or the like is provided, a transistor having a light-transmitting property, such as an organic transistor using an organic semiconductor material or a transistor using an oxide semiconductor, is preferably used.

[0217] A display region **5202** is a display device provided in a pillar portion into which the light-emitting device described in Embodiment 1 is incorporated. The display region **5202** can compensate for the view hindered by the pillar by displaying an image taken by an imaging means provided on the car body. Similarly, the display region **5203** provided in the dashboard portion can compensate for the view hindered by the car body by displaying an image taken by an imaging means provided on the outside of the automobile. Thus, blind areas can be compensated for and the safety can be enhanced. Showing an image so as to compensate for the area that cannot be seen makes it possible to confirm safety more naturally and comfortably.

[0218] The display region **5203** can provide a variety of kinds of information by displaying navigation data, a speedometer, a tachometer, a mileage, a fuel meter, a gearshift state, air-condition setting, and the like. The content or layout of the display can be changed as appropriate in accordance with the preference of a user. Note that such information can also be provided on the display region **5200** to the display region **5202**. The display region **5200** to the display region **5203** can also be used as lighting devices.

[0219] FIGS. **12(A)** and **12(B)** illustrate a foldable portable information terminal **5150**. The foldable portable information terminal **5150** includes a housing **5151**, a display region **5152**, and a bend portion **5153**. FIG. **12(A)** illustrates the portable information terminal **5150** that is opened.

[0220] FIG. **12(B)** illustrates the portable information terminal **5150** that is folded. The portable information terminal **5150** is compact in size and has excellent portability when folded, despite its

large display region **5152**.

[0221] The display region **5152** can be folded in half with the bend portion **5153**. The bend portion **5153** includes a flexible member and a plurality of supporting members, and when the display region is folded, the flexible member expands and the bend portion **5153** has a radius of curvature of 2 mm or more, preferably 3 mm or more.

[0222] Note that the display region **5152** may be a touch panel (an input/output device) including a touch sensor (an input device). The light-emitting apparatus of one embodiment of the present invention can be used for the display region **5152**.

[0223] FIGS. **13(A)** to **13(C)** illustrate a foldable portable information terminal **9310**. FIG. **13(A)** illustrates the portable information terminal **9310** that is opened. FIG. **13(B)** illustrates the portable information terminal **9310** which is in the state of being changed from one of an opened state and a folded state to the other. FIG. **13(C)** illustrates the portable information terminal **9310** that is folded. The portable information terminal **9310** is excellent in portability when folded, and is excellent in display browsability when opened because of a seamless large display region.

[0224] A display panel **9311** is supported by three housings **9315** joined together by hinges **9313**. Note that the display panel **9311** may be a touch panel (an input/output device) including a touch sensor (an input device). By folding the display panel **9311** at the hinges **9313** between two housings **9315**, the portable information terminal **9310** can be reversibly changed in shape from the opened state to the folded state. A light-emitting apparatus of one embodiment of the present invention can be used for the display panel **9311**. A display region **9312** in the display panel **9311** is a display region that is positioned at a side surface of the portable information terminal **9310** which is folded.

Example 1

[0225] In this example, the behavior of external quantum efficiency and a blue index (BI) in light-emitting devices having microcavity structures with varied optical path lengths will be described in detail. Structures of organic compounds used in a light-emitting device **1-1** to a light-emitting device **1-8** fabricated in this example are shown below.

##STR00006## ##STR00007##

<Method for Fabricating Light-Emitting Device **1-1** to Light-Emitting Device **1-8**>

[0226] First, over a glass substrate, an alloy film of silver (Ag), palladium (Pd), and copper (Cu) (an Ag—Pd—Cu (APC) film) was formed as a reflective electrode to a thickness of 100 nm by a sputtering method, and a film of indium tin oxide containing silicon oxide (ITSO) was formed as a transparent electrode to a thickness of 85 nm by a sputtering method, so that the first electrode **101** was formed. Note that the area of the electrode was 4 mm^{sup.2} (2 mm×2 mm).

[0227] Next, in pretreatment for forming the light-emitting device over the substrate, the surface of the substrate was washed with water and baked at 200° C. for one hour, and then UV ozone treatment was performed for 370 seconds.

[0228] After that, the substrate was transferred into a vacuum evaporation apparatus in which the pressure was reduced to about 10^{sup.-4} Pa, vacuum baking at 170° C. for 30 minutes was performed in a heating chamber in the vacuum evaporation apparatus, and then the substrate was naturally cooled down for about 30 minutes.

[0229] Next, the substrate on which the first electrode **101** was formed was fixed to a substrate holder provided in the vacuum evaporation apparatus such that the side on which the first electrode **101** was formed faced downward, and 3-[4-(9-phenanthryl)-phenyl]-9-phenyl-9H-carbazole (abbreviation: PCPPn) represented as the above structural formula (i) and molybdenum(VI) oxide were deposited by co-evaporation on the first electrode **101** to have a weight ratio of 1:0.5 (=PCPPn: molybdenum oxide) by an evaporation method, whereby the hole-injection layer **111** was formed.

[0230] Variation of the optical path length in the light-emitting devices was achieved by changing the thickness of the hole-injection layer **111**. In the light-emitting device **1-1**, the thickness of the

hole-injection layer **111** was 10 nm; in the light-emitting device **1-2**, that was 15 nm; in the light-emitting device **1-3**, that was 20 nm; in the light-emitting device **1-4**, that was 25 nm; in the light-emitting device **1-5**, that was 30 nm; in the light-emitting device **1-6**, that was 35 nm; in the light-emitting device **1-7**, that was 40 nm; and in the light-emitting device **1-8**, that was 45 nm.

[0231] Next, on the hole-injection layer **111**, PCPPn was deposited by evaporation to a thickness of 15 nm to form the hole-transport layer **112**.

[0232] Subsequently, 7-[4-(10-phenyl-9-anthryl)phenyl]-7H-dibenzo[c,g]carbazole (abbreviation: cgDBCzPA) expressed by the above structural formula (ii) and N,N'-diphenyl-N,N'-bis(3-methylphenyl)-N,N'-bis[3-(9-phenyl-9H-fluoren-9-yl)phenyl]-pyrene-1,6-diamine (abbreviation: 1.6 mMemFLPAPrn) expressed by the above structural formula (iii) were deposited by co-evaporation to a thickness of 25 nm at a weight ratio of 1:0.03 (=cgDBCzPA: 1.6 mMemFLPAPrn), so that the light-emitting layer **113** was formed.

[0233] After that, on the light-emitting layer **113**, cgDBCzPA was deposited by evaporation to a thickness of 5 nm, and 2,9-bis(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline (abbreviation: NBPhen) expressed by the above structural formula (iv) was deposited by evaporation to a thickness of 15 nm, whereby the electron-transport layer **114** was formed.

[0234] After the formation of the electron-transport layer **114**, lithium fluoride (LiF) was deposited by evaporation to a thickness of 1 nm to form the electron-injection layer **115**, and silver (Ag) and magnesium (Mg) in a volume ratio of 1:0.1 were deposited by evaporation to a thickness of 10 nm to form the second electrode **102**, whereby the light-emitting device **1-1** to the light-emitting device **1-8** were fabricated. Note that the second electrode **102** is a semi-transmissive and semi-reflective electrode having a function of reflecting light and a function of transmitting light, and the light-emitting devices of this example are top-emission devices from which light is extracted through thesecond electrode **102**. In addition, 1,3,5-t (dibenzothi ophen-4-yl)-benzene (abbreviation: DBT3P-II) expressed by the above structural formula (v) was deposited by evaporation to a thickness of 70 nm on thesecond electrode **102**, whereby theextraction efficiency was improved.

[0235] The device structures of the light-emitting device **1-1** to the light-emitting device **1-8** are listed in the following Table 1 and Table 2. Table 2 shows the thickness of the hole-injection layer in each light-emitting device.

TABLE-US-00001

Layer	Thickness (nm)
Organic cap layer DBT3P-II	70
Semi-transmissive and Ag:Mg	10
nm semi-reflective electrode (1:0.1)	
Electron-injection layer LiF	1
Electron-transport layer NBPhen	15
cgDBCzPA	5
Light-emitting layer cgDBCzPA:1,6mFLPAPrn	25
(1:0.03)	
Hole-transport layer PCPPn	15
Hole-injection layer PCPPn:MoOx	x
(1:0.5)	
Transparent electrode ITSO	85
Reflective electrode APC	100

TABLE-US-00002

Device	Thickness (nm)
Light-emitting device 1-1	10
Light-emitting device 1-2	15
Light-emitting device 1-3	20
Light-emitting device 1-4	25
Light-emitting device 1-5	30
Light-emitting device 1-6	35
Light-emitting device 1-7	40
Light-emitting device 1-8	45

[0236] Each of the fabricated light-emitting devices was sealed using a glass substrate (a sealant was applied to surround the device, followed by UV treatment and one-hour heat treatment at 80° C. at the time of sealing) in a glove box containing a nitrogen atmosphere so that the light-emitting device is not exposed to the air, and then measurements were carried out. Note that the measurement was performed at room temperature.

[0237] Measurement results of the light-emitting device **1-1** to the light-emitting device **1-8** are shown below. All measurement results are shown as values at luminance around 1000 cd/m. In addition, the external quantum efficiency is uncorrected external quantum efficiency calculated from front luminance on the assumption of Lambertian distribution.

TABLE-US-00003

Device	Current (mA)	External quantum efficiency (%)	EL_max (cd/A)	Chromaticity x	Chromaticity y	BI (nm)
Light-emitting device 1-1	0.139	0.050	0.813	1.46	16.1	459
Light-emitting device 1-2	0.139	0.055	1.58	2.71	29.0	459
Light-emitting device 1-3	0.138	0.059	2.84	4.70	48.4	459
Light-emitting device 1-4	0.139	0.062	4.55	7.27	73.1	460

emitting device 1-5 0.137 0.068 6.28 9.59 93.0 460 Light-emitting device 1-6 0.137 0.079 8.64 12.2 110 463 Light-emitting device 1-7 0.136 0.093 10.3 13.3 111 465 Light-emitting device 1-8 0.133 0.122 12.4 13.8 102 468

[0238] FIG. 14 shows a PL spectrum of 1,6mFL PA Prn that is an emission center substance in a toluene solution. For measurement of the PL spectrum, a fluorescence spectrophotometer (Edinburgh Instruments FS920 manufactured by Hamamatsu Photonics K.K.) was used, and in calculation of the average photon energy $E_{\text{sub.ave}}$, the formula (1) shown in Embodiment 1 was used. By this, the average photon energy ($E_{\text{sub.ave}}$) of the PL spectrum of 1,6mFLPA Prn in a toluene solution was calculated to be 2.61 eV. Furthermore, the emission edge on the short wavelength side of the PL spectrum of 1,6mFLPA Prn in the toluene solution was calculated in a manner shown in FIG. 15 to be 438 nm, and its energy was 2.83 eV.

[0239] FIG. 16 is a graph showing a relation of external quantum efficiency (EQE) and a blue index (BI) with respect to the emission peak energy ($E_{\text{sub.em}}$) of each light-emitting device. The horizontal axis represents $E_{\text{sub.em}}$, and the vertical axes represent EQE and BI.

[0240] According to the diagram, both EQE and BI exhibit favorable values in the light-emitting device having a cavity length enabling a value of $E_{\text{sub.em}}$ to fall within a range between $E_{\text{sub.ave}}$ (2.61 eV in this example) and 2.69 eV.

[0241] Note that a value of 2.69 eV is the energy corresponding to 0.95 times $E_{\text{sub.edge}}$ as 2.83 eV.

[0242] As described above, it was found that the blue light-emitting device with a microcavity structure can emit light with favorable efficiency when the light-emitting device has a cavity length enabling a wavelength corresponding to energy that is higher than or equal to $E_{\text{sub.ave}}$ and less than or equal to 0.95 times $E_{\text{sub.edge}}$. In other words, the cavity length is controlled to satisfy $E_{\text{sub.ave}} \leq E_{\text{sub.em}} \leq 0.95E_{\text{sub.edge}}$ (Range A in the drawing), whereby both EQE and BI can be maximized.

[0243] In particular, it was found that the light-emitting device whose $E_{\text{sub.em}}$ is greater than or equal to $E_{\text{sub.ave}}+0.02$ (eV) and less than or equal to $E_{\text{sub.edge}} \times 0.95$ (eV) is preferable in consideration of BI; thus, the cavity length is preferably controlled to satisfy $E_{\text{sub.ave}}+0.02 \leq E_{\text{sub.em}} \leq 0.95E_{\text{sub.edge}}$ (Range B in the drawing).

Example 2

[0244] In this example, the behavior of external quantum efficiency and blue index (BI) in light-emitting devices with microcavity structures with varied optical path lengths will be described in detail. Structures of organic compounds used in a light-emitting device 2-1 to a light-emitting device 2-8 fabricated in this example are shown below.

##STR00008## ##STR00009##

<Method for Fabricating Light-Emitting Device 2-1 to Light-Emitting Device 2-8>

[0245] First, over a glass substrate, an alloy film of silver (Ag), palladium (Pd), and copper (Cu) (an Ag—Pd—Cu (APC) film) was formed as a reflective electrode to a thickness of 100 nm by a sputtering method, and a film of indium tin oxide containing silicon oxide (ITSO) was formed as a transparent electrode to a thickness of 85 nm by a sputtering method, so that the first electrode 101 was formed. Note that the area of the electrode was 4 mm^{sup.2} (2 mm×2 mm).

[0246] Next, in pretreatment for forming the light-emitting device over the substrate, the surface of the substrate was washed with water and baked at 200° C. for one hour, and then UV ozone treatment was performed for 370 seconds.

[0247] After that, the substrate was transferred into a vacuum evaporation apparatus in which the pressure was reduced to about 10^{sup.-4} Pa, vacuum baking at 170° C. for 30 minutes was performed in a heating chamber in the vacuum evaporation apparatus, and then the substrate was naturally cooled down for about 30 minutes.

[0248] Next, the substrate on which the first electrode 101 was formed was fixed to a substrate holder provided in the vacuum evaporation apparatus such that the side on which the first electrode

101 was formed faced downward, and 3-[4-(9-phenanthryl)-phenyl]-9-phenyl-9H-carbazole (abbreviation: PCPPn) expressed by the above structural formula (i) and molybdenum(VI) oxide were deposited by co-evaporation on the first electrode **101** to have a weight ratio of 1:0.5 (=PCPPn: molybdenum oxide) by an evaporation method, whereby the hole-injection layer **111** was formed.

[0249] Variation of the optical path length in the light-emitting devices was achieved by changing the thickness of the hole-injection layer **111**. In the light-emitting device **2-1**, the thickness of the hole-injection layer **111** was 10 nm; in the light-emitting device **2-2**, that was 15 nm; in the light-emitting device **2-3**, that was 20 nm; in the light-emitting device **2-4**, that was 25 nm; in the light-emitting device **2-5**, that was 30 nm; in the light-emitting device **2-6**, that was 35 nm; in light-emitting device **2-7**, that was 40 nm; and in the light-emitting device **2-8**, that was 45 nm.

[0250] Next, on the hole-injection layer **111**, PCPPn was deposited by evaporation to a thickness of 15 nm to form the hole-transport layer **112**.

[0251] Subsequently, 7-[4-(10-phenyl-9-anthryl)phenyl]-7H-dibenzo[c,g]carbazole (abbreviation: cgDBCzPA) expressed by the above structural formula (ii) and N,N'-bis(3-methylphenyl)-N,N'-bis[3-(9-phenyl-9H-fluoren-9-yl)phenyl]-pyrene-1,6-diamine (abbreviation: 1,6mMemFLPAPrn) expressed by the above structural formula (vi) were deposited by co-evaporation to a thickness of 25 nm at a weight ratio of 1:0.03 (=cgDBCzPA: 1.6 mMemFLPAPrn), so that the light-emitting layer **113** was formed.

[0252] After that, on the light-emitting layer **113**, cgDBCzPA was deposited by evaporation to a thickness of 5 nm, and 2,9-bis(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline (abbreviation: NBPhen) expressed by the above structural formula (iv) was deposited by evaporation to a thickness of 15 nm, whereby the electron-transport layer **114** was formed.

[0253] After the formation of the electron-transport layer **114**, lithium fluoride (LiF) was deposited by evaporation to a thickness of 1 nm to form the electron-injection layer **115**, and silver (Ag) and magnesium (Mg) in a volume ratio of 1:0.1 were deposited by evaporation to a thickness of 15 nm to form the second electrode **102**, whereby the light-emitting device **2-1** to the light-emitting device **2-8** were fabricated. Note that the second electrode **102** is a semi-transmissive and semi-reflective electrode having a function of reflecting light and a function of transmitting light, and the light-emitting devices of this example are top-emission devices from which light is extracted through the second electrode **102**. In addition, 1,3,5-tri (dibenzothiophen-4-yl)-benzene (abbreviation: DBT3P-II) expressed by the above structural formula (v) was deposited by evaporation to a thickness of 70 nm on the second electrode **102**, whereby the extraction efficiency was improved.

[0254] The device structures of the light-emitting device **2-1** to the light-emitting device **2-8** are listed in the following Table 4 and Table 5. Table 5 shows the thickness of the hole-injection layer in each light-emitting device.

TABLE-US-00004 TABLE 4 Organic cap layer DBT3P-II 70 nm Semi-transmissive and Ag:Mg 15 nm semi-reflective (1:0.1) electrode Electron-injection layer LiF 1 nm Electron-transport layer NBPhen 15 nm cgDBCzPA 5 nm Light-emitting layer cgDBCzPA:1,6mMemFLPAPrn 25 nm (1:0.03) Hole-transport layer PCPPn 15 nm Hole-injection layer PCPPn:MoOx x nm (1:0.5) Transparent electrode ITSO 85 nm Reflective electrode APC 100 nm

TABLE-US-00005 TABLE 5 Thickness (nm) Light-emitting device 2-1 10 Light-emitting device 2-2 15 Light-emitting device 2-3 20 Light-emitting device 2-4 25 Light-emitting device 2-5 30 Light-emitting device 2-6 35 Light-emitting device 2-7 40 Light-emitting device 2-8 45

[0255] Each of the fabricated light-emitting devices was sealed using a glass substrate (a sealant was applied to surround the device, followed by UV treatment and one-hour heat treatment at 80° C. at the time of sealing) in a glove box containing a nitrogen atmosphere so that the light-emitting device is not exposed to the air, and then measurements were carried out. Note that the measurement was performed at room temperature.

[0256] Measurement results of the light-emitting device **2-1** to the light-emitting device **2-8** are

shown below. All measurement results are represented as values at luminance around 1000 cd/m. In addition, the external quantum efficiency is uncorrected quantum efficiency calculated from front luminance on the assumption of Lambertian distribution.

TABLE-US-00006 TABLE 6 External Current quantum efficiency efficiency EL_max

Chromaticity x	Chromaticity y (cd/A)	(%)	BI (nm)	Light-emitting device	2-1	0.143	0.062	0.522
0.84	8.49	461	Light-emitting device 2-2	0.137	0.064	1.02	1.60	15.9
462	Light-emitting device 2-3	0.138	0.068	1.96	2.99	28.9	461	Light-emitting device 2-4
0.137	0.069	3.26	4.84	46.9	462	Light-emitting device 2-5	0.138	0.074
4.86	7.04	65.7	462	Light-emitting device 2-6	0.137	0.081	6.97	
9.66	85.7	463	Light-emitting device 2-7	0.135	0.098	9.66	12.2	99.0
466	Light-emitting device 2-8	0.132	0.125	12.0	13.2	95.5	469	

[0257] FIG. 17 shows a PL spectrum of 1.6 mMemFLPAPrn that is an emission center substance in a toluene solution. For measurement of the PL spectrum, a fluorescence spectrophotometer (Edinburgh Instruments FS920 manufactured by Hamamatsu Photonics K.K.) was used, and in calculation of the average photon energy E.sub.ave, the formula (1) shown in Embodiment 1 was used. By this, the average photon energy (E.sub.ave) of the PL spectrum of 1.6 mMemFLPAPrn in a toluene solution was calculated to be 2.59 eV. Furthermore, the emission edge on the short wavelength side of the PL spectrum of 1.6 mMemFLPAPrn in the toluene solution was calculated in a manner shown in FIG. 18 to be 440 nm, and its energy was 2.82 eV.

[0258] FIG. 19 is a graph showing a relation of external quantum efficiency (EQE) and blue index (BI) with respect to the emission peak energy (E.sub.em) of each light-emitting device. The horizontal axis represents E.sub.em, and the vertical axes represent EQE and BI.

[0259] According to the diagram, both EQE and BI exhibit favorable values in the light-emitting device having a cavity length enabling a value of E.sub.em to fall within a range between E.sub.ave (2.59 eV in this example) and 2.68 eV.

[0260] Note that a value of 2.68 eV is the energy corresponding to 0.95 times E.sub.edge as 2.82 eV.

[0261] As described above, it was found that the blue light-emitting device with a microcavity structure can emit light with favorable efficiency when the light-emitting device has a cavity length enabling a wavelength corresponding to energy that is higher than or equal to E.sub.ave and less than or equal to 0.95 times E.sub.edge. In other words, the cavity length is controlled to satisfy $E_{sub.ave} \leq E_{sub.em} \leq 0.95 E_{sub.edge}$ (Range A in the drawing), whereby both EQE and BI can be maximized.

[0262] In particular, it was found that the light-emitting device whose E.sub.em is greater than or equal to $E_{sub.ave} + 0.02$ (eV) and less than or equal to $E_{sub.edge} \times 0.95$ (eV) is preferable in consideration of BI; thus, the cavity length is preferably controlled to satisfy $E_{sub.ave} + 0.02 \leq E_{sub.em} \leq 0.95 E_{sub.edge}$ (Range B in the drawing).

Example 3

[0263] In this example, the behavior of external quantum efficiency and blue index (BI) in light-emitting devices with microcavity structures with varied optical path lengths will be described in detail. Structures of organic compounds used in a light-emitting device 3-1 to a light-emitting device 3-8 fabricated in this example are shown below.

##STR00010## ##STR00011##

[0264] <Method for Fabricating Light-Emitting Device 3-1 to Light-Emitting Device 3-8>

[0265] First, over a glass substrate, an alloy film of silver (Ag), palladium (Pd), and copper (Cu) (an Ag—Pd—Cu (APC) film) was formed as a reflective electrode to a thickness of 100 nm by a sputtering method, and a film of indium tin oxide containing silicon oxide (ITSO) was formed as a transparent electrode to a thickness of 95 nm by a sputtering method, so that the first electrode 101 was formed. Note that the area of the electrode was 4 mm² (2 mm×2 mm).

[0266] Next, in pretreatment for forming the light-emitting device over the substrate, the surface of the substrate was washed with water and baked at 200° C. for one hour, and then UV ozone

treatment was performed for 370 seconds.

[0267] After that, the substrate was transferred into a vacuum evaporation apparatus in which the pressure was reduced to about 10.sup.-4 Pa, vacuum baking at 170° C. for 30 minutes was performed in a heating chamber in the vacuum evaporation apparatus, and then the substrate was naturally cooled down for about 30 minutes.

[0268] Next, the substrate on which the first electrode **101** was formed was fixed to a substrate holder provided in the vacuum evaporation apparatus such that the side on which the first electrode **101** was formed faced downward, and 3-[4-(9-phenanthryl)-phenyl]-9-phenyl-9H-carbazole (abbreviation: PCPPn) expressed by the above structural formula (i) and molybdenum(VI) oxide were deposited by co-evaporation on the first electrode **101** to have a weight ratio of 1:0.5 (=PCPPn: molybdenum oxide) by an evaporation method, whereby the hole-injection layer **111** was formed.

[0269] Variation of the optical path length in the light-emitting devices was achieved by changing the thickness of the hole-injection layer **111**. In the light-emitting device **3-1**, the thickness of the hole-injection layer **111** was 10 nm; in the light-emitting device **3-2**, that was 15 nm; in the light-emitting device **3-3**, that was 20 nm; in the light-emitting device **3-4**, that was 25 nm; in the light-emitting device **3-5**, that was 30 nm; in the light-emitting device **3-6**, that was 35 nm; in the light-emitting device **3-7**, that was 40 nm; and in the light-emitting device **3-8**, that was 45 nm.

[0270] Next, on the hole-injection layer **111**, PCPPn was deposited by evaporation to a thickness of 15 nm to form the hole-transport layer **112**.

[0271] Subsequently, 7-[4-(10-phenyl-9-anthryl)phenyl]-7H-dibenzo[c,g]carbazole (abbreviation: cgDBCzPA) expressed by the structural formula (ii) and N,N'-(pyrene-1,6-diyl)bis[(6,N-diphenylbenzo[b]naphtho[1,2-d]furan)-8-amine](abbreviation: 1,6BnfAPrn-03) expressed by the structural formula (vii) were deposited to a thickness of 25 nm by co-evaporation at a weight ratio of 1:0.03 (=cgDBCzPA:1,6BnfAPrn-03), whereby the light-emitting layer **113** was formed.

[0272] After that, on the light-emitting layer **113**, cgDBCzPA was deposited by evaporation to a thickness of 5 nm, and 2,9-bis(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline (abbreviation: NBPhen) expressed by the above structural formula (iv) was deposited by evaporation to a thickness of 15 nm, whereby the electron-transport layer **114** was formed.

[0273] After the formation of the electron-transport layer **114**, lithium fluoride (LiF) was deposited by evaporation to a thickness of 1 nm to form the electron-injection layer **115**, and silver (Ag) and magnesium (Mg) in a volume ratio of 1:0.1 were deposited by evaporation to a thickness of 15 nm to form the second electrode **102**, whereby the light-emitting device **3-1** to the light-emitting device **3-8** were fabricated. Note that the second electrode **102** is a semi-transmissive and semi-reflective electrode having a function of reflecting light and a function of transmitting light, and the light-emitting devices of this example are top-emission devices from which light is extracted through the second electrode **102**. In addition, 1,3,5-tri (dibenzothiophen-4-yl)-benzene (abbreviation: DBT3P-II) expressed by the above structural formula (v) was deposited by evaporation to a thickness of 70 nm on the second electrode **102**, whereby the extraction efficiency was improved.

[0274] The device structures of the light-emitting device **3-1** to the light-emitting device **3-8** are listed in the following Table 7 and Table 8. Table 8 shows the thickness of the hole-injection layer in each light-emitting device.

TABLE-US-00007 TABLE 7 Organic cap layer DBT3P-II 70 nm Semi-transmissive and Ag:Mg 15 nm semi-reflective (1:0.1) electrode Electron-injection layer LiF 1 nm Electron-transport layer NBPhen 15 nm cgDBCzPA 5 nm Light-emitting layer cgDBCzPA:1,6BnfAPrn-03 25 nm (1:0.03) Hole-transport layer PCPPn 15 nm Hole-injection layer PCPPn:MoOx x nm (1:0.5) Transparent electrode ITSO 95 nm Reflective electrode APC 100 nm

TABLE-US-00008 TABLE 8 Thickness (nm) Light-emitting device 3-1 10 Light-emitting device 3-2 15 Light-emitting device 3-3 20 Light-emitting device 3-4 25 Light-emitting device 3-5 30 Light-emitting device 3-6 35 Light-emitting device 3-7 40 Light-emitting device 3-8 45

[0275] Each of the fabricated light-emitting devices was sealed using a glass substrate (a sealant was applied to surround the device, followed by UV treatment and one-hour heat treatment at 80° C. at the time of sealing) in a glove box containing a nitrogen atmosphere so that the light-emitting device is not exposed to the air, and then measurements were carried out. Note that the measurement was performed at room temperature.

[0276] Measurement results of the light-emitting device 3-1 to the light-emitting device 3-8 are shown below. All measurement results are represented as values at luminance around 1000 cd/m. In addition, the external quantum efficiency is uncorrected quantum efficiency calculated from front luminance on the assumption of Lambertian distribution.

TABLE-US-00009 TABLE 9 Current External quantum efficiency efficiency EL_max Chromaticity x Chromaticity y (cd/A) (%) BI (nm) Light-emitting device 3-1 0.143 0.030 0.344 0.800 11.6 452 Light-emitting device 3-2 0.142 0.030 0.69 1.60 22.8 452 Light-emitting device 3-3 0.142 0.032 1.15 2.63 36.2 453 Light-emitting device 3-4 0.142 0.033 2.01 4.51 60.5 453 Light-emitting device 3-5 0.142 0.036 3.25 6.95 90.1 455 Light-emitting device 3-6 0.141 0.046 4.83 9.30 105 457 Light-emitting device 3-7 0.139 0.061 6.19 10.3 102 460 Light-emitting device 3-8 0.136 0.087 7.71 10.6 89.1 465

[0277] FIG. 20 shows a PL spectrum of 1,6BnfAPrn-03 that is an emission center substance in a toluene solution. For measurement of the PL spectrum, a fluorescence spectrophotometer (Edinburgh Instruments FS920 manufactured by Hamamatsu Photonics K.K.) was used, and in calculation of the average photon energy E.sub.ave, the formula (1) shown in Embodiment 1 was used. By this, the average photon energy (E.sub.ave) of the PL spectrum of 1,6BnfAPrn-03 in a toluene solution was calculated to be 2.65 eV. Furthermore, the emission edge on the short wavelength side of the PL spectrum was calculated in a manner shown in FIG. 21 to be 432 nm, and its energy was 2.87 eV.

[0278] FIG. 22 is a graph showing a relation of external quantum efficiency (EQE) and blue index (BI) with respect to the emission peak energy (E.sub.em) of each light-emitting device. The horizontal axis represents E.sub.em, and the vertical axes represent EQE and BI.

[0279] According to the diagram, both EQE and BI exhibit favorable values in the light-emitting device having a cavity length enabling a value of E.sub.em to fall within a range between E.sub.ave (2.65 eV in this example) and 2.73 eV.

[0280] Note that a value of 2.73 eV is the energy corresponding to 0.95 times E.sub.edge as 2.87 eV.

[0281] As described above, it was found that the blue light-emitting device with a microcavity structure can emit light with favorable efficiency when the light-emitting device has a cavity length enabling a wavelength corresponding to energy that is higher than or equal to E.sub.ave and less than or equal to 0.95 times E.sub.edge. In other words, the cavity length is controlled to satisfy $E_{sub.ave} \leq E_{sub.em} \leq 0.95 E_{sub.edge}$ (Range A in the drawing), whereby both EQE and BI can be maximized.

[0282] In particular, it was found that the light-emitting device whose E.sub.em is greater than or equal to $E_{sub.ave} + 0.02$ (eV) (2.67 eV in this example) and less than or equal to $E_{sub.edge} \times 0.95$ (eV) is preferable in consideration of BI; thus, the cavity length is preferably controlled to satisfy $E_{sub.ave} + 0.02 \leq E_{sub.em} \leq 0.95 E_{sub.edge}$ (Range B in the drawing).

Example 4

[0283] In this example, the behavior of external quantum efficiency and blue index (BI) in light-emitting devices with microcavity structures with varied optical path lengths will be described in detail. Structures of organic compounds used in a light-emitting device 4-1 to a light-emitting device 4-4 fabricated in this example are shown below.

##STR00012## ##STR00013##

<Method for Fabricating Light-Emitting Device 4-1 to Light-Emitting Device 4-4>

[0284] First, over a glass substrate, a film of indium tin oxide containing silicon oxide (ITSO) was

formed as a transparent electrode to a thickness of 70 nm, an alloy film of silver (Ag), palladium (Pd), and copper (Cu) (an Ag—Pd—Cu (APC) film) was formed as a semi-transmissive and semi-reflective electrode to a thickness of 25 nm by a sputtering method, and then a film of ITSO was formed as a transparent electrode to a thickness of 10 nm by a sputtering method, so that the first electrode **101** was formed. Note that the area of the electrode was 4 mm² (2 mm×2 mm). [0285] Next, in pretreatment for forming the light-emitting device over the substrate, the surface of the substrate was washed with water and baked at 200° C. for one hour, and then UV ozone treatment was performed for 370 seconds.

[0286] After that, the substrate was transferred into a vacuum evaporation apparatus in which the pressure was reduced to about 10⁻⁴ Pa, vacuum baking at 170° C. for 30 minutes was performed in a heating chamber in the vacuum evaporation apparatus, and then the substrate was naturally cooled down for about 30 minutes.

[0287] Next, the substrate on which the first electrode **101** was formed was fixed to a substrate holder provided in the vacuum evaporation apparatus such that the side on which the first electrode **101** was formed faced downward, and 3-[4-(9-phenanthryl)-phenyl]-9-phenyl-9H-carbazole (abbreviation: PCPPn) expressed by the above structural formula (i) and molybdenum(VI) oxide were deposited by co-evaporation on the first electrode **101** to have a weight ratio of 1:0.5 (=PCPPn: molybdenum oxide) by an evaporation method, whereby the hole-injection layer **111** was formed.

[0288] Variation of the optical path length in the light-emitting devices was achieved by changing the thickness of the hole-injection layer **111**. In the light-emitting device **4-1**, the thickness of the hole-injection layer **111** was 10 nm; in the light-emitting device **4-2**, that was 12.5 nm; in the light-emitting device **4-3**, that was 15 nm; and in the light-emitting device **4-4**, that was 17.5 nm.

[0289] Next, on the hole-injection layer **111**, PCPPn was deposited by evaporation to a thickness of 15 nm to form the hole-transport layer **112**.

[0290] Subsequently, 7-[4-(10-phenyl-9-anthryl)phenyl]-7H-dibenzo[c,g]carbazole (abbreviation: cgDBCzPA) expressed by the structural formula (ii) and N,N'-(pyrene-1,6-diyl)bis[(6,N-diphenylbenzo[b]naphtho[1,2-d]furan)-8-amine] (abbreviation: 1,6BnfAPrn-03) expressed by the structural formula (vii) were deposited to a thickness of 20 nm by co-evaporation at a weight ratio of 1:0.03 (=cgDBCzPA:1,6BnfAPrn-03), whereby the light-emitting layer **113** was formed.

[0291] After that, over the light-emitting layer **113**, 2-[3'-(dibenzothiophen-4-yl)biphenyl-3-yl]dibenzo[f,h]quinoxaline (abbreviation: 2mDBTBPDQ-II) expressed by the structural formula (viii) was deposited by evaporation to a thickness of 10 nm, and 2,9-bis(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline (abbreviation: NBPhen) expressed by the structural formula (iv) was deposited by evaporation to a thickness of 10 nm, whereby the electron-transport layer **114** was formed.

[0292] After the formation of the electron-transport layer **114**, lithium fluoride (LiF) was deposited by evaporation to a thickness of 1 nm to form the electron-injection layer **115**, and then aluminum was deposited by evaporation to a thickness of 120 nm to form the second electrode **102**, whereby the light-emitting device **4-1** to the light-emitting device **4-4** of this example were fabricated. Note that the second electrode **102** is an electrode that reflects light, and the light-emitting devices of this example are bottom-emission devices from which light is extracted through the first electrode **101**.

[0293] The device structures of the light-emitting device **4-1** to the light-emitting device **4-4** are listed in Table 10 and Table 11. Table 11 shows the thickness of the hole-injection layer in each light-emitting device.

TABLE-US-00010 TABLE 10 Organic cap layer Al 120 nm Semi-transmissive and LiF 1 nm semi-reflective electrode Electron-injection layer NBPhen 10 nm Electron-transport layer 2mDBTBPDQ-II 10 nm cgDBCzPA:1,6BnfAPrn-03 20 nm (1:0.03) Light-emitting layer PCPPn 15 nm Hole-transport layer PCPPn:MoO_x x nm (1:0.5) Hole-injection layer ITSO 10 nm Transparent electrode APC 25 nm Reflective electrode ITSO 70 nm

TABLE-US-00011 TABLE 11 Thickness (nm) Light-emitting device 4-1 10 Light-emitting device 4-2 12.5 Light-emitting device 4-3 15 Light-emitting device 4-4 17.5

[0294] Each of the fabricated light-emitting devices was sealed using a glass substrate (a sealant was applied to surround the device, followed by UV treatment and one-hour heat treatment at 80° C. at the time of sealing) in a glove box containing a nitrogen atmosphere so that the light-emitting device is not exposed to the air, and then measurements were carried out. Note that the measurement was performed at room temperature.

[0295] Measurement results of the light-emitting device 4-1 to the light-emitting device 4-4 are shown below. All measurement results are represented as values at luminance around 1000 cd/m. In addition, the external quantum efficiency is uncorrected quantum efficiency calculated from front luminance on the assumption of Lambertian distribution.

TABLE-US-00012 TABLE 12 External Current quantum efficiency efficiency EL_max Chromaticity x Chromaticity y (cd/A) (%) BI (nm) Light-emitting device 4-1 0.141 0.039 4.13 8.15 105 453 Light-emitting device 4-2 0.140 0.044 5.55 10.6 127 455 Light-emitting device 4-3 0.139 0.052 6.66 11.6 129 459 Light-emitting device 4-4 0.138 0.074 8.41 12.5 114 462

[0296] FIG. 23 shows a PL spectrum of 1,6BnfAPrn-03 that is an emission center substance in a toluene solution. For measurement of the PL spectrum, a fluorescence spectrophotometer (Edinburgh Instruments FS920 manufactured by Hamamatsu Photonics K.K.) was used, and in calculation of the average photon energy E.sub.ave, the formula (1) shown in Embodiment 1 was used. By this, the average photon energy (E.sub.ave) of the PL spectrum of 1,6BnfAPrn-03 in the toluene solution was calculated to be 2.65 eV. Furthermore, the emission edge on the short wavelength side of the PL spectrum was calculated in a manner shown in FIG. 24 to be 432 nm, and its energy was 2.87 eV.

[0297] FIG. 25 is a graph showing a relation of external quantum efficiency (EQE) and blue index (BI) with respect to the emission peak energy (E.sub.em) of each light-emitting device. The horizontal axis represents E.sub.em, and the vertical axes represent EQE and BI.

[0298] According to the diagram, both EQE and BI exhibit favorable values in the light-emitting device having a cavity length enabling a value of E.sub.em to fall within a range between E.sub.ave (2.65 eV in this example) and 2.73 eV.

[0299] Note that a value of 2.73 eV is the energy corresponding to 0.95 times E.sub.edge as 2.87 eV.

[0300] As described above, it was found that the blue light-emitting device with a microcavity structure can emit light with favorable efficiency when the light-emitting device has a cavity length enabling a wavelength corresponding to energy that is higher than or equal to E.sub.ave and less than or equal to 0.95 times E.sub.edge. In other words, the cavity length is controlled to satisfy E.sub.ave ≤ E.sub.em ≤ 0.95E.sub.edge (Range A in the drawing), whereby both EQE and BI can be maximized.

[0301] In particular, it was found that the light-emitting device whose E.sub.em is greater than or equal to E.sub.ave+0.02 (eV) (2.67 eV in this example) and less than or equal to E.sub.edge×0.95 (eV) is preferable in consideration of BI; thus, the cavity length is preferably controlled to satisfy E.sub.ave+0.02 ≤ E.sub.em ≤ 0.95E.sub.edge (Range B in the drawing).

Example 5

[0302] In this example, the behavior of external quantum efficiency and blue index (BI) in light-emitting devices with microcavity structures with varied optical path lengths will be described in detail. Structures of organic compounds used in a light-emitting device 5-1 to a light-emitting device 5-8 fabricated in this example are shown below.

##STR00014## ##STR00015##

<Method for Fabricating Light-Emitting Device 5-1 to Light-Emitting Device 5-8>

[0303] First, over a glass substrate, an alloy film of silver (Ag), palladium (Pd), and copper (Cu) (an Ag—Pd—Cu (APC) film) was formed as a reflective electrode to a thickness of 100 nm by a

sputtering method, and a film of indium tin oxide containing silicon oxide (ITSO) was formed as a transparent electrode to a thickness of 85 nm by a sputtering method, so that the first electrode **101** was formed. Note that the area of the electrode was 4 mm^{sup.2} (2 mm×2 mm).

[0304] Next, in pretreatment for forming the light-emitting device over the substrate, the surface of the substrate was washed with water and baked at 200° C. for one hour, and then UV ozone treatment was performed for 370 seconds.

[0305] After that, the substrate was transferred into a vacuum evaporation apparatus in which the pressure was reduced to about 10^{sup.-4} Pa, vacuum baking at 170° C. for 30 minutes was performed in a heating chamber in the vacuum evaporation apparatus, and then the substrate was naturally cooled down for about 30 minutes.

[0306] Next, the substrate on which the first electrode **101** was formed was fixed to a substrate holder provided in the vacuum evaporation apparatus such that the side on which the first electrode **101** was formed faced downward, and 3-[4-(9-phenanthryl)-phenyl]-9-phenyl-9H-carbazole (abbreviation: PCPPn) expressed by the above structural formula (i) and molybdenum(VI) oxide were deposited by co-evaporation on the first electrode **101** to have a weight ratio of 1:0.5 (=PCPPn: molybdenum oxide) by an evaporation method, whereby the hole-injection layer **111** was formed.

[0307] Variation of the optical path length in the light-emitting devices was achieved by changing the thickness of the hole-injection layer **111**. In the light-emitting device **5-1**, the thickness of the hole-injection layer **111** was 10 nm; in the light-emitting device **5-2**, that was 15 nm; in the light-emitting device **5-3**, that was 20 nm; in the light-emitting device **5-4**, that was 25 nm; in the light-emitting device **5-5**, that was 30 nm; in the light-emitting device **5-6**, that was 35 nm; in the light-emitting device **5-7**, that was 40 nm; and in the light-emitting device **5-8**, that was 45 nm.

[0308] Next, on the hole-injection layer **111**, PCPPn was deposited by evaporation to a thickness of 15 nm to form the hole-transport layer **112**.

[0309] Subsequently, 7-[4-(10-phenyl-9-anthryl)phenyl]-7H-dibenzo[c,g]carbazole (abbreviation: cgDBCzPA) expressed by the structural formula (ii) and N,N'-(pyrene-1,6-diyl)bis(N-phenyl-6-cyclohexylbenzo[b]naphtho[1,2-d]furan-8-amine) (abbreviation: 1,6chBnfAPrn) expressed by the above structural formula (ix) were deposited to a thickness of 25 nm by co-evaporation at a weight ratio of 1:0.03 (=cgDBCzPA:1,6chBnfAPrn), whereby the light-emitting layer **113** was formed.

[0310] After that, on the light-emitting layer **113**, cgDBCzPA was deposited by evaporation to a thickness of 5 nm, and 2,9-bis(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline (abbreviation: NBPhen) expressed by the above structural formula (iv) was deposited by evaporation to a thickness of 15 nm, whereby the electron-transport layer **114** was formed.

[0311] After the formation of the electron-transport layer **114**, lithium fluoride (LiF) was deposited by evaporation to a thickness of 1 nm to form the electron-injection layer **115**, and silver (Ag) and magnesium (Mg) in a volume ratio of 1:0.1 were deposited by evaporation to a thickness of 10 nm to form the second electrode **102**, whereby the light-emitting device **5-1** to the light-emitting device **5-8** were fabricated. Note that the second electrode **102** is a semi-transmissive and semi-reflective electrode having a function of reflecting light and a function of transmitting light, and the light-emitting devices of this example are top-emission devices from which light is extracted through the second electrode **102**. In addition, 1,3,5-tri (dibenzothiophen-4-yl)-benzene (abbreviation: DBT3P-II) expressed by the above structural formula (v) was deposited by evaporation to a thickness of 70 nm on the second electrode **102**, whereby the extraction efficiency was improved.

[0312] The device structures of the light-emitting device **5-1** to the light-emitting device **5-8** are listed in Table 13 and Table 14. Table 14 shows the thickness of the hole-injection layer in each light-emitting device.

TABLE-US-00013 TABLE 13 Organic cap layer DBT3P-II 70 nm Semi-transmissive and Ag:Mg 10 nm semi-reflective (1:0.1) electrode Electron-injection layer LiF 1 nm Electron-transport layer NBPhen 15 nm cgDBCzPA 5 nm Light-emitting layer cgDBCzPA:1,6chBnfAPrn 25 nm (1:0.03)

Hole-transport layer PCPPn 15 nm Hole-injection layer PCPPn:MoOx x nm (1:0.5) Transparent electrode ITSO 85 nm Reflective electrode APC 100 nm

TABLE-US-00014 TABLE 14 Thickness (nm) Light-emitting device 5-1 10 Light-emitting device 5-2 15 Light-emitting device 5-3 20 Light-emitting device 5-4 25 Light-emitting device 5-5 30 Light-emitting device 5-6 35 Light-emitting device 5-7 40 Light-emitting device 5-8 45

[0313] Each of the fabricated light-emitting devices was sealed using a glass substrate (a sealant was applied to surround the device, followed by UV treatment and one-hour heat treatment at 80° C. at the time of sealing) in a glove box containing a nitrogen atmosphere so that the light-emitting device is not exposed to the air, and then measurements were carried out. Note that the measurement was performed at room temperature.

[0314] Measurement results of the light-emitting device 5-1 to the light-emitting device 5-8 are shown below. All measurement results are represented as values at luminance around 1000 cd/m.sup.2. In addition, the external quantum efficiency is uncorrected quantum efficiency calculated from front luminance on the assumption of Lambertian distribution.

TABLE-US-00015 TABLE 15 External Current quantum efficiency efficiency EL_max Chromaticity x Chromaticity y (cd/A) (%) BI (nm) Light-emitting device 5-1 0.142 0.032 0.735 1.70 23.3 453 Light-emitting device 5-2 0.142 0.034 1.39 3.13 41.6 453 Light-emitting device 5-3 0.142 0.035 2.28 4.93 65.0 452 Light-emitting device 5-4 0.141 0.040 3.50 7.15 88.5 453 Light-emitting device 5-5 0.140 0.045 4.86 9.22 109 455 Light-emitting device 5-6 0.138 0.058 6.72 11.2 117 458 Light-emitting device 5-7 0.138 0.078 8.32 11.9 107 460 Light-emitting device 5-8 0.135 0.103 9.52 11.6 92.1 463

[0315] FIG. 26 shows a PL spectrum of 1,6chBnfAPrn that is an emission center substance in a toluene solution. For measurement of the PL spectrum, a fluorescence spectrophotometer (Edinburgh Instruments FS920 manufactured by Hamamatsu Photonics K.K.) was used, and in calculation of the average photon energy E.sub.ave, the formula (1) shown in Embodiment 1 was used. By this, the average photon energy (E.sub.ave) of the PL spectrum of 1,6chBnfAPrn in the toluene solution was calculated to be 2.65 eV. Furthermore, the emission edge on the short wavelength side of the PL spectrum was calculated in a manner shown in FIG. 27 to be 432 nm, and its energy was 2.87 eV.

[0316] FIG. 28 is a graph showing a relation of external quantum efficiency (EQE) and blue index (BI) with respect to the emission peak energy (E.sub.em) of each light-emitting device. The horizontal axis represents E.sub.em, and the vertical axes represent EQE and BI.

[0317] According to the diagram, both EQE and BI exhibit favorable values in the light-emitting device having a cavity length enabling a value of E.sub.em to fall within a range between E.sub.ave (2.65 eV in this example) and 2.73 eV.

[0318] Note that a value of 2.73 eV is the energy corresponding to 0.95 times E.sub.edge as 2.87 eV.

[0319] As described above, it was found that the blue light-emitting device with a microcavity structure can emit light with favorable efficiency when the light-emitting device has a cavity length enabling a wavelength corresponding to energy that is higher than or equal to E.sub.ave and less than or equal to 0.95 times E.sub.edge. In other words, the cavity length is controlled to satisfy E.sub.ave ≤ E.sub.em ≤ 0.95E.sub.edge (Range A in the drawing), whereby both EQE and BI can be maximized.

[0320] In particular, it was found that the light-emitting device whose E.sub.em is greater than or equal to E.sub.ave+0.02 (eV) (2.67 eV in this example) and less than or equal to E.sub.edge×0.95 (eV) is preferable in consideration of BI; thus, the cavity length is preferably controlled to satisfy E.sub.ave+0.02 ≤ E.sub.em ≤ 0.95E.sub.edge (Range B in the drawing).

Example 6

[0321] In this example, the behavior of external quantum efficiency and blue index (BI) in light-emitting devices with microcavity structures with varied optical path lengths will be described in

detail. Structures of organic compounds used in a light-emitting device **6-1** to a light-emitting device **6-8** fabricated in this example are shown below.

##STR00016##

<Method for Fabricating Light-Emitting Device **6-1** to Light-Emitting Device **6-8**>

[0322] First, over a glass substrate, an alloy film of silver (Ag), palladium (Pd), and copper (Cu) (an Ag—Pd—Cu (APC) film) was formed as a reflective electrode to a thickness of 100 nm by a sputtering method, and a film of indium tin oxide containing silicon oxide (ITSO) was formed as a transparent electrode to a thickness of 85 nm by a sputtering method, so that the first electrode **101** was formed. Note that the area of the electrode was 4 mm^{sup.2} (2 mm×2 mm).

[0323] Next, in pretreatment for forming the light-emitting device over the substrate, the surface of the substrate was washed with water and baked at 200° C. for one hour, and then UV ozone treatment was performed for 370 seconds.

[0324] After that, the substrate was transferred into a vacuum evaporation apparatus in which the pressure was reduced to about 10^{sup.-4} Pa, vacuum baking at 170° C. for 30 minutes was performed in a heating chamber in the vacuum evaporation apparatus, and then the substrate was naturally cooled down for about 30 minutes.

[0325] Next, the substrate on which the first electrode **101** was formed was fixed to a substrate holder provided in the vacuum evaporation apparatus such that the side on which the first electrode **101** was formed faced downward, and 3-[4-(9-phenanthryl)-phenyl]-9-phenyl-9H-carbazole (abbreviation: PCPPn) expressed by the above structural formula (i) and molybdenum(VI) oxide were deposited by co-evaporation on the first electrode **101** to have a weight ratio of 1:0.5 (=PCPPn: molybdenum oxide) by an evaporation method, whereby the hole-injection layer **111** was formed.

[0326] Variation of the optical path length in the light-emitting devices was achieved by changing the thickness of the hole-injection layer **111**. In the light-emitting device **6-1**, the thickness of the hole-injection layer **111** was 10 nm; in the light-emitting device **6-2**, that was 15 nm; in the light-emitting device **6-3**, that was 20 nm; in the light-emitting device **6-4**, that was 25 nm; in the light-emitting device **6-5**, that was 30 nm; in the light-emitting device **6-6**, that was 35 nm; in the light-emitting device **6-7**, that was 40 nm; and in the light-emitting device **6-8**, that was 45 nm.

[0327] Next, on the hole-injection layer **111**, PCPPn was deposited by evaporation to a thickness of 15 nm to form the hole-transport layer **112**.

[0328] Subsequently, 7-[4-(10-phenyl-9-anthryl)phenyl]-7H-dibenzo[c,g]carbazole (abbreviation: cgDBCzPA) expressed by the above structural formula (ii) and 3,10-bis[N-(9-phenyl-9H-carbazol-2-yl)-N-phenylamino]naphtho[2,3-b;6,7-b']bisbenzofuran (abbreviation: 3,10PCA2Nbf(IV)-02) expressed by the above structural formula (x) were deposited by co-evaporation to have a weight ratio of 1:0.01 (=cgDBCzPA: 3,10PCA2Nbf(IV)-02) to a thickness of 25 nm, whereby the light-emitting layer **113** was formed.

[0329] After that, on the light-emitting layer **113**, cgDBCzPA was deposited by evaporation to a thickness of 5 nm, and 2,9-bis(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline (abbreviation: NBPhen) expressed by the above structural formula (iv) was deposited by evaporation to a thickness of 15 nm, whereby the electron-transport layer **114** was formed.

[0330] After the formation of the electron-transport layer **114**, lithium fluoride (LiF) was deposited by evaporation to a thickness of 1 nm to form the electron-injection layer **115**, and silver (Ag) and magnesium (Mg) in a volume ratio of 1:0.1 (=Ag:Mg) were deposited by evaporation to a thickness of 10 nm to form the second electrode **102**, whereby the light-emitting device **6-1** to the light-emitting device **6-8** were fabricated. Note that the second electrode **102** is a semi-transmissive and semi-reflective electrode having a function of reflecting light and a function of transmitting light, and the light-emitting devices of this example are top-emission devices from which light is extracted through the second electrode **102**. In addition, 1,3,5-tri(dibenzothiophen-4-yl)-benzene (abbreviation: DBT3P-II) expressed by the above structural formula (v) was deposited by

evaporation to a thickness of 70 nm on the second electrode **102**, whereby the extraction efficiency was improved.

[0331] The device structures of the light-emitting device **6-1** to the light-emitting device **6-8** are listed in Table 16 and Table 17. Table 17 shows the thickness of the hole-injection layer in each light-emitting device.

TABLE-US-00016 TABLE 16 Organic cap layer DBT3P-II 70 nm Semi-transmissive and Ag:Mg 10 nm semi-reflective (1:0.1) electrode Electron-injection layer LiF 1 nm Electron-transport layer NBPhen 15 nm cgDBCzPA 5 nm Light-emitting layer cgDBCzPA:3,10PCA2Nbf(IV)-02 25 nm (1:0.01) Hole-transport layer PCPPn 15 nm Hole-injection layer PCPPn:MoOx x nm (1:0.5) Transparent electrode ITSO 85 nm Reflective electrode APC 100 nm

TABLE-US-00017 TABLE 17 Thickness (nm) Light-emitting device 6-1 10 Light-emitting device 6-2 15 Light-emitting device 6-3 20 Light-emitting device 6-4 25 Light-emitting device 6-5 30 Light-emitting device 6-6 35 Light-emitting device 6-7 40 Light-emitting device 6-8 45

[0332] Each of the fabricated light-emitting devices was sealed using a glass substrate (a sealant was applied to surround the device, followed by UV treatment and one-hour heat treatment at 80° C. at the time of sealing) in a glove box containing a nitrogen atmosphere so that the light-emitting device is not exposed to the air, and then measurements were carried out. Note that the measurement was performed at room temperature.

[0333] Measurement results of the light-emitting device **6-1** to the light-emitting device **6-8** are shown below. All measurement results are represented as values at luminance around 1000 cd/in.sup.2. In addition, the external quantum efficiency is uncorrected quantum efficiency calculated from front luminance on the assumption of Lambertian distribution.

TABLE-US-00018 TABLE 18 Current External quantum efficiency efficiency EL_max Chromaticity x Chromaticity y (cd/A) (%) BI (nm) Light-emitting device 6-1 0.145 0.036 0.433 0.897 11.9 454 Light-emitting device 6-2 0.142 0.036 0.857 1.84 23.7 454 Light-emitting device 6-3 0.141 0.039 1.63 3.42 42.0 454 Light-emitting device 6-4 0.141 0.041 2.68 5.42 65.8 454 Light-emitting device 6-5 0.140 0.044 4.07 7.93 92.8 455 Light-emitting device 6-6 0.140 0.049 5.41 9.84 111 455 Light-emitting device 6-7 0.139 0.056 7.16 11.9 127 458 Light-emitting device 6-8 0.138 0.078 8.80 12.7 114 460

[0334] FIG. **29** shows a PL spectrum of 3,10PCA2Nbf(IV)-02 that is an emission center substance in a toluene solution. For measurement of the PL spectrum, an absolute PL quantum yield measurement system (Quantaurs-QY manufactured by Hamamatsu Photonics K.K.) was used, and in calculation of the average photon energy E.sub.ave, the formula (VII) shown in Embodiment 1 was used because the vertical axis of the PL spectrum shown in FIG. **29** represents a normalized spectrum I(A) proportional to energy $\phi_{sub.p}(\lambda)$. By this, the average photon energy (E.sub.ave) of the PL spectrum of 3,10PCA2Nbf(IV)-02 in the toluene solution was calculated to be 2.66 eV. Furthermore, the emission edge on the short wavelength side of the PL spectrum was calculated in a manner shown in FIG. **30** to be 431 nm, and its energy was 2.88 eV.

[0335] FIG. **31** is a graph showing a relation of external quantum efficiency (EQE) and blue index (BI) with respect to the emission peak energy (E.sub.em) of each light-emitting device. The horizontal axis represents E.sub.em, and the vertical axes represent EQE and BI.

[0336] According to the diagram, both EQE and BI exhibit favorable values in the light-emitting device having a cavity length enabling a value of E.sub.em to fall within a range between E.sub.ave (2.66 eV in this example) and 2.73 eV.

[0337] Note that a value of 2.73 eV is the energy corresponding to 0.95 times E.sub.edge as 2.88 eV.

[0338] As described above, it was found that the blue light-emitting device with a microcavity structure can emit light with favorable efficiency when the light-emitting device has a cavity length enabling a wavelength corresponding to energy that is higher than or equal to E.sub.ave and less than or equal to 0.95 times E.sub.edge. In other words, the cavity length is controlled to satisfy

E.sub.ave E.sub.em 0.95E.sub.edge (Range A in the drawing), whereby both EQE and BI can be maximized.

[0339] In particular, it was found that the light-emitting device whose E.sub.em is greater than or equal to E.sub.ave+0.02 (eV) (2.67 eV in this example) and less than or equal to E.sub.edge×0.95 (eV) is preferable in consideration of BI; thus, the cavity length is preferably controlled to satisfy E.sub.ave+0.02 E.sub.em 0.95E.sub.edge (Range B in the drawing).

Example 7

[0340] In this example, the behavior of external quantum efficiency and blue index (BI) in light-emitting devices with microcavity structures with varied optical path lengths will be described in detail. Structures of organic compounds used in a light-emitting device **7-1** to a light-emitting device **7-8** fabricated in this example are shown below.

##STR00017##

<Method for Fabricating Light-Emitting Device **7-1** to Light-Emitting Device **7-8**>

[0341] First, over a glass substrate, an alloy film of silver (Ag), palladium (Pd), and copper (Cu) (an Ag—Pd—Cu (APC) film) was formed as a reflective electrode to a thickness of 100 nm by a sputtering method, and a film of indium tin oxide containing silicon oxide (ITSO) was formed as a transparent electrode to a thickness of 85 nm by a sputtering method, so that the first electrode **101** was formed. Note that the area of the electrode was 4 mm^{sup.2} (2 mm×2 mm).

[0342] Next, in pretreatment for forming the light-emitting device over the substrate, the surface of the substrate was washed with water and baked at 200° C. for one hour, and then UV ozone treatment was performed for 370 seconds.

[0343] After that, the substrate was transferred into a vacuum evaporation apparatus in which the pressure was reduced to about 10.sup.-4 Pa, vacuum baking at 170° C. for 30 minutes was performed in a heating chamber in the vacuum evaporation apparatus, and then the substrate was naturally cooled down for about 30 minutes.

[0344] Next, the substrate on which the first electrode **101** was formed was fixed to a substrate holder provided in the vacuum evaporation apparatus such that the side on which the first electrode **101** was formed faced downward, and 3-[4-(9-phenanthryl)-phenyl]-9-phenyl-9H-carbazole (abbreviation: PCPPn) expressed by the above structural formula (i) and molybdenum(VI) oxide were deposited by co-evaporation on the first electrode **101** to have a weight ratio of 1:0.5 (=PCPPn: molybdenum oxide) by an evaporation method, whereby the hole-injection layer **111** was formed.

[0345] Variation of the optical path length in the light-emitting devices was achieved by changing the thickness of the hole-injection layer **111**. In the light-emitting device **7-1**, the thickness of the hole-injection layer **111** was 10 nm; in the light-emitting device **7-2**, that was 15 nm; in the light-emitting device **7-3**, that was 20 nm; in the light-emitting device **7-4**, that was 25 nm; in the light-emitting device **7-5**, that was 30 nm; in the light-emitting device **7-6**, that was 35 nm; in the light-emitting device **7-7**, that was 40 nm; and in the light-emitting device **7-8**, that was 45 nm.

[0346] Next, on the hole-injection layer **111**, PCPPn was deposited by evaporation to a thickness of 15 nm to form the hole-transport layer **112**.

[0347] Subsequently, 7-[4-(10-phenyl-9-anthryl)phenyl]-7H-dibenzo[c,g]carbazole (abbreviation: cgDBCzPA) expressed by the above structural formula (ii) and 3,10-bis[N-(dibenzofuran-3-yl)-N-phenylamino]naphtho[2,3-b;6,7-b']bisbenzofuran (abbreviation: 3,10FrA2Nbf(IV)-02) expressed by the above structural formula (xi) were deposited by co-evaporation to have a weight ratio of 1:0.01 (=cgDBCzPA: 3,10FrA2Nbf(IV)-02) to a thickness of 25 nm, whereby the light-emitting layer **113** was formed.

[0348] After that, on the light-emitting layer **113**, cgDBCzPA was deposited by evaporation to a thickness of 5 nm, and 2,9-bis(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline (abbreviation: NBPhen) expressed by the above structural formula (iv) was deposited by evaporation to a thickness of 15 nm, whereby the electron-transport layer **114** was formed.

[0349] After the formation of the electron-transport layer **114**, lithium fluoride (LiF) was deposited by evaporation to a thickness of 1 nm to form the electron-injection layer **115**, and silver (Ag) and magnesium (Mg) in a volume ratio of 1:0.1 (=Ag:Mg) were deposited by evaporation to a thickness of 10 nm to form the second electrode **102**, whereby the light-emitting device **7-1** to the light-emitting device **7-8** were fabricated. Note that the second electrode **102** is a semi-transmissive and semi-reflective electrode having a function of reflecting light and a function of transmitting light, and the light-emitting device of this example are top-emission devices from which light is extracted through the second electrode **102**. In addition, 1,3,5-tri(dibenzothiophen-4-yl)-benzene (abbreviation: DBT3P-II) expressed by the above structural formula (v) was deposited by evaporation to a thickness of 70 nm on the second electrode **102**, whereby the extraction efficiency was improved.

[0350] The device structures of the light-emitting device **7-1** to the light-emitting device **7-8** are listed in Table 19 and Table 20. Table 20 shows the thickness of the hole-injection layer in each light-emitting device.

TABLE-US-00019 TABLE 19 Organig cap layer DBT3P-II 70 nm Semi-transmissive and Ag:Mg 10 nm semi-reflective (1:0.1) electrode Electron-injection layer LiF 1 nm Electron-transport layer NBPhen 15 nm cgDBCzPA 5 nm Light-emitting layer cgDBCzPA:3,10FrA2Nbf(IV)-02 25 nm (1:0.01) Hole-transport layer PCPPn 15 nm Hole-injection layer PCPPn:MoOx x nm (1:0.5) Transparent electrode ITSO 85 nm Reflective electrode APC 100 nm

TABLE-US-00020 TABLE 20 Thickness (nm) Light-emitting device 7-1 10 Light-emitting device 7-2 15 Light-emitting device 7-3 20 Light-emitting device 7-4 25 Light-emitting device 7-5 30 Light-emitting device 7-6 35 Light-emitting device 7-7 40 Light-emitting device 7-8 45

[0351] Each of the fabricated light-emitting devices was sealed using a glass substrate (a sealant was applied to surround the device, followed by UV treatment and one-hour heat treatment at 80° C. at the time of sealing) in a glove box containing a nitrogen atmosphere so that the light-emitting device is not exposed to the air, and then measurements were carried out. Note that the measurement was performed at room temperature.

[0352] Measurement results of the light-emitting device **7-1** to the light-emitting device **7-8** are shown below. All measurement results are represented as values at luminance around 1000 cd/in.sup.2. In addition, the external quantum efficiency is uncorrected quantum efficiency calculated from front luminance on the assumption of Lambertian distribution.

TABLE-US-00021 TABLE 21 Current External quantum efficiency efficiency EL_max Chromaticity x Chromaticity y (cd/A) (%) BI (nm) Light-emitting device 7-1 0.145 0.024 0.382 1.06 16.2 448 Light-emitting device 7-2 0.143 0.025 0.72 1.95 29.2 448 Light-emitting device 7-3 0.143 0.026 1.38 3.64 54.3 449 Light-emitting device 7-4 0.143 0.027 2.18 5.50 80.4 449 Light-emitting device 7-5 0.142 0.031 3.37 8.01 109 450 Light-emitting device 7-6 0.141 0.036 4.47 9.68 123 451 Light-emitting device 7-7 0.140 0.047 5.73 10.8 122 453 Light-emitting device 7-8 0.139 0.064 7.09 11.4 111 455

[0353] FIG. **32** shows a PL spectrum of 3,10PCA2Nbf(IV)-02 that is an emission center substance in a toluene solution. For measurement of the PL spectrum, an absolute PL quantum yield measurement system (Quantaaurus-QY manufactured by Hamamatsu Photonics K.K.) was used, and in calculation of the average photon energy E.sub.ave, the formula (VII) shown in Embodiment 1 was used because the vertical axis of the PL spectrum shown in FIG. **32** represents a normalized spectrum I(A) proportional to energy $\phi_{sub.p}(\lambda)$. By this, the average photon energy (E.sub.ave) of the PL spectrum of 3,10PCA2Nbf(IV)-02 in the toluene solution was calculated to be 2.71 eV. Furthermore, the emission edge on the short wavelength side of the PL spectrum was calculated in a manner shown in FIG. **33** to be 426 nm, and its energy was 2.91 eV.

[0354] FIG. **34** is a graph showing a relation of external quantum efficiency (EQE) and blue index (BI) with respect to the emission peak energy (E.sub.em) of each light-emitting device. The horizontal axis represents E.sub.em, and the vertical axes represent EQE and BI.

[0355] According to the diagram, both EQE and BI exhibit favorable values in the light-emitting device having a cavity length enabling a value of $E_{sub.em}$ to fall within a range between $E_{sub.ave}$ (2.71 eV in this example) and 2.76 eV.

[0356] Note that a value of 2.76 eV is the energy corresponding to 0.95 times $E_{sub.edge}$ as 2.91 eV.

[0357] As described above, it was found that the blue light-emitting device with a microcavity structure can emit light with favorable efficiency when the light-emitting device has a cavity length enabling a wavelength corresponding to energy that is higher than or equal to $E_{sub.ave}$ and less than or equal to 0.95 times $E_{sub.edge}$. In other words, the cavity length is controlled to satisfy $E_{sub.ave} \leq E_{sub.em} \leq 0.95E_{sub.edge}$ (Range A in the drawing), whereby both EQE and BI can be maximized.

[0358] In particular, it was found that the light-emitting device whose $E_{sub.em}$ is greater than or equal to $E_{sub.ave} + 0.02$ (eV) (2.73 eV in this example) and less than or equal to $E_{sub.edge} \times 0.95$ (eV) is preferable in consideration of BI; thus, the cavity length is preferably controlled to satisfy $E_{sub.ave} + 0.02 \leq E_{sub.em} \leq 0.95E_{sub.edge}$ (Range B in the drawing).

REFERENCE NUMERALS

[0359] **101**: first electrode, **101-1**: reflective electrode, **101-2**: transparent electrode, **102**: second electrode, **103**: EL layer, **104**: organic cap layer, **111**: hole-injection layer, **112**: hole-transport layer, **113**: light-emitting layer, **114**: electron-transport layer, **115**: electron-injection layer, **116**: charge-generation layer, **117**: P-type layer, **118**: electron-relay layer, **119**: electron-injection buffer layer, **400**: substrate, **401**: first electrode, **403**: EL layer, **404**: second electrode, **405**: sealant, **406**: sealant, **407**: sealing substrate, **412**: pad, **420**: IC chip, **601**: driver circuit portion (source line driver circuit), **602**: pixel portion, **603**: driver circuit portion (gate line driver circuit), **604**: sealing substrate, **605**: sealant, **607**: space, **608**: wiring, **609**: FPC (flexible printed circuit), **610**: element substrate, **611**: switching FET, **612**: current control FET, **613**: first electrode, **614**: insulator, **616**: EL layer, **617**: second electrode, **618**: light-emitting device, **2001**: housing, **2002**: light source, **2100**: robot, **2110**: arithmetic device, **2101**: illuminance sensor, **2102**: microphone, **2103**: upper camera, **2104**: speaker, **2105**: display, **2106**: lower camera, **2107**: obstacle sensor, **2108**: moving mechanism, **3001**: lighting device, **5000**: housing, **5001**: display portion, **5002**: second display portion, **5003**: speaker, **5004**: LED lamp, **5005**: operation key, **5006**: connection terminal, **5007**: sensor, **5008**: microphone, **5012**: support, **5013**: earphone, **5100**: cleaning robot, **5101**: display, **5102**: camera, **5103**: brush, **5104**: operation button, **5150**: information terminal, **5151**: housing, **5152**: display region, **5153**: bend portion, **5120**: dust, **5200**: display region, **5201**: display region, **5202**: display region, **5203**: display region, **7101**: housing, **7103**: display portion, **7105**: stand, **7107**: display portion, **7109**: operation key, **7110**: remote controller, **7201**: main body, **7202**: housing, **7203**: display portion, **7204**: keyboard, **7205**: external connection port, **7206**: pointing device, **7210**: second display portion, **7401**: housing, **7402**: display portion, **7403**: operation button, **7404**: external connection port, **7405**: speaker, **7406**: microphone, **7400**: mobile phone, **9310**: portable information terminal, **9311**: display panel, **9312**: display region, **9313**: hinge, **9315**: housing

Claims

1. A light-emitting device comprising: a first electrode; a second electrode; and an EL layer between the first electrode and the second electrode, the EL layer comprising an emission center substance, wherein the first electrode and the second electrode overlap each other, wherein one of the first electrode and the second electrode is a reflective electrode and the other is a semi-transmissive and semi-reflective electrode, wherein $E_{sub.ave}$ denotes an average of photon energy [eV] of an emission spectrum of the emission center substance, wherein $E_{sub.edge}$ denotes an energy [eV] at an emission edge on a short wavelength side of the emission spectrum, wherein

E.sub.em denotes a photon energy [eV] at a peak wavelength of light emitted from the light-emitting device, wherein an energy [eV] at the peak wavelength of the emission spectrum is higher than 0.95E_{edge}, wherein a difference between E.sub.em and E.sub.ave is within 0.08 eV, and wherein the following formula (1) is satisfied: $E_{ave} \leq E_{em} \leq 0.95E_{edge}$. (1)

2. The light-emitting device according to claim 1, wherein the emission spectrum of the emission center substance is an emission spectrum obtained from a solution comprising the emission center substance.

3. The light-emitting device according to claim 2, wherein a solvent of the solution has a dielectric constant greater than or equal to 1 and less than or equal to 10 at room temperature.

4. The light-emitting device according to claim 2, wherein a solvent of the solution is toluene or chloroform.

5. The light-emitting device according to claim 1, wherein the emission center substance is capable of emitting blue light.

6. The light-emitting device according to claim 1, wherein E.sub.em is greater than or equal to 2.6 eV and less than or equal to 2.9 eV.

7. The light-emitting device according to claim 1, wherein a difference between E.sub.em and E.sub.ave is 0.02 eV or more.

8. The light-emitting device according to claim 1, wherein a blue index of the light-emitting device is 89.1 or more.

9. The light-emitting device according to claim 1, wherein the emission spectrum of the emission center substance is obtained by counting photons emitted from the emission center substance per wavelength, wherein E.sub.ave is estimated by the following formula (IV):

$$E_{ave} = \frac{\phi_p}{N_p} = ch \cdot \text{Math.} \frac{\int F(\lambda) / d}{\int F(\lambda) d}, \quad (\text{IV}) \quad \text{wherein } \phi_{sub.p} \text{ is defined as a total energy [eV] of}$$

light emission of the emission center substance, wherein N_p is defined as a total number of photons of the light emission of the emission center substance, wherein Δ is a functional variable representing wavelength of a light spectrum, wherein c is light speed, wherein h is the Planck constant, and wherein F(λ) represents a function of the emission spectrum of the emission center substance.

10. The light-emitting device according to claim 1, wherein the emission spectrum of the emission center substance is obtained by measuring energy itself emitted from the emission center substance or by normalized spectrum that is proportional to the energy, wherein E.sub.ave is estimated by the

following formula (VI): $E_{ave} = \frac{\phi_p}{N_p} = ch \cdot \text{Math.} \frac{\int \phi_p(\lambda) d}{\int \phi_p(\lambda) d}, \quad (\text{VI})$ wherein $\phi_{sub.p}$ is defined as a

total energy [eV] of light emission of the emission center substance, wherein N_p is defined as a total number of photons of the light emission of the emission center substance, wherein λ is a functional variable representing wavelength of a light spectrum, wherein c is light speed, wherein h is the Planck constant, and wherein $\phi_{sub.p}(\lambda)$ corresponds to a light emission energy or a proportional value to the light emission energy.

11. The light-emitting device according to claim 1, further comprising an organic compound layer on a first surface of the semi-transmissive and semi-reflective electrode, wherein the organic compound layer comprises an organic compound with molecular weight greater than or equal to 300 and less than or equal to 1200, wherein a second surface of the semi-transmissive and semi-reflective electrode and the reflective electrode face each other, and wherein the second surface is an opposite surface to the first surface.

12. A light-emitting apparatus comprising: the light-emitting device according to claim 1, and one of a transistor and a substrate.

13. An electronic device comprising: the light-emitting apparatus according to claim 12, and one of

a sensor, an operation button, a speaker and a microphone.

14. A lighting device comprising: the light-emitting apparatus according to claim **12**, and a housing.
